Dissecting 5G in the Wild: Performance, Coverage, and Support for Next-Gen Applications

Moinak Ghoshal
Ph.D. Candidate

Northeastern University
Institute for Intelligent Networked Systems

5G's Promise

- Initial 5G rollout in 2019



Ultra-High Bandwidth

Multi-Gbps data rates for massive data exchange



Ultra-Low Latency

Sub-10 millisecond-level response times for real-time applications

5G's Promise

- Potential to unlock 5G "killer apps"

Fully immersive digital environments

Overlaying digital information on real-world environments



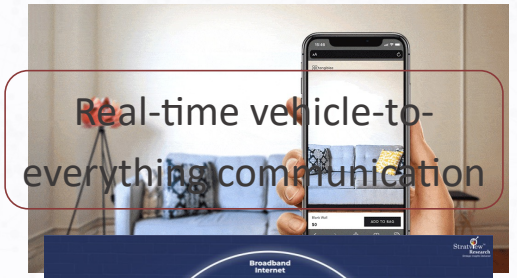
Virtual Reality

Augmented Reality

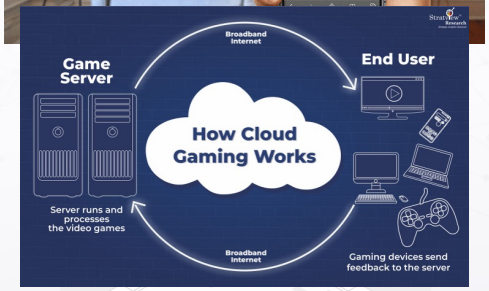
Connected Autonomous Vehicles

360˚/Volumetric Video Streaming

Cloud Gaming



Real-time vehicle-to-everything communication



Panoramic video for immersive experiences

Console-quality gaming on any device

5G “Killer Apps”

Cloud/Edge-Assisted Processing

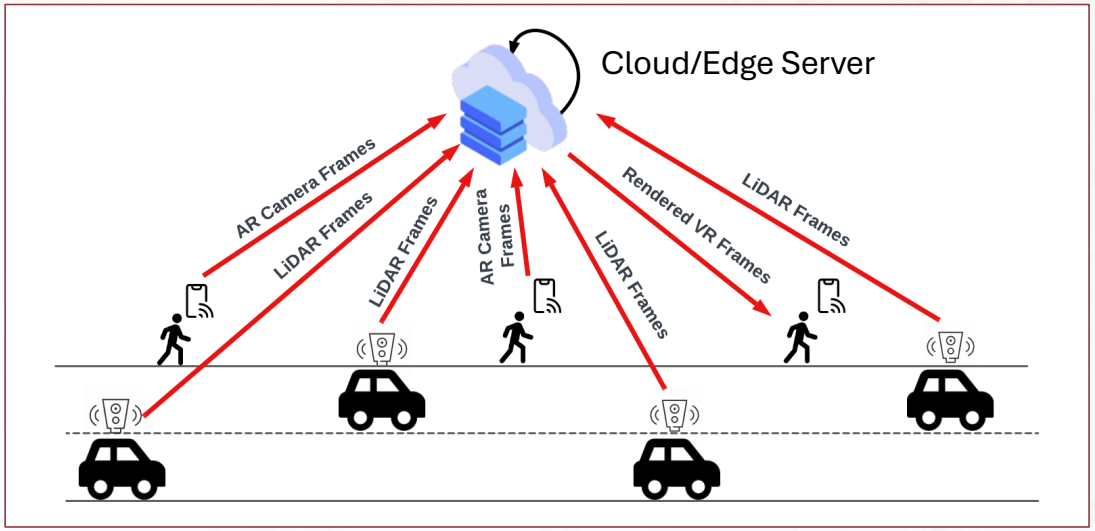
Powerful servers handle SLAM, object detection, frame rendering

Large Data Volume Exchange

Upload camera/LIDAR frames, download rendered frames

Stringent Quality of Experience (QoE) Requirements

High frame rates, ultra-low motion-to-photon latency



5G “killer apps” are **bandwidth-hungry** and **latency-critical**

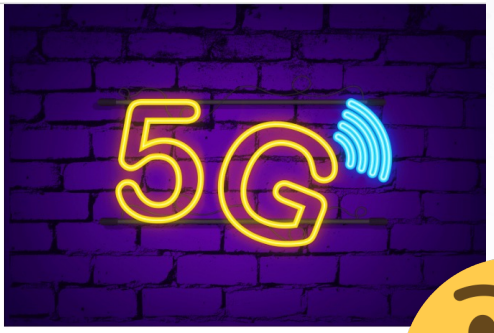
Does Today's 5G Fulfill its Promise?

THE WALL STREET JOURNAL.

PERSONAL TECHNOLOGY: JOANNA STERN

It's Not Just You: 5G Is a Big Letdown

The newest network won't change how you use your phone—but you can still make the most of it



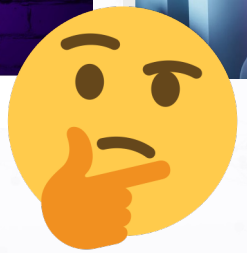
ELENA SCOTTI/THE WALL STREET JOURNAL, SHUTTERSTOCK; ISTOCK

5G Networks Are Performing Worse. What's Going On? › Millimeter-wave fizzles and regulatory hurdles plague progress

BY MICHAEL KOZIOL | 06 MAY 2023 | 4 MIN READ |

Michael Koziol is the news manager at IEEE Spectrum.



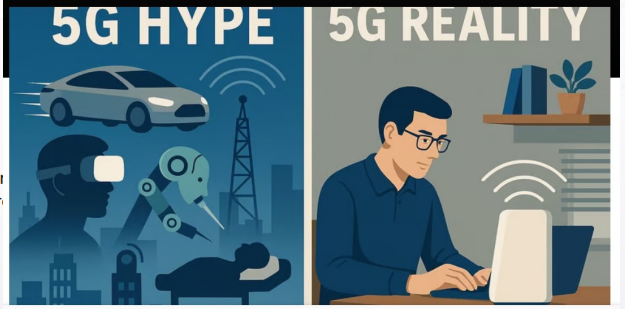


AI

Remember 5G hype? Here's what's actually happening now

By Mitch Wagner · Apr 14, 2025 10:50am

5G | fixed wireless access | network slicing | Standalone 5G (5G SA)



FEBRUARY 28, 2023 · Editors' notes

How 5G disappointed 'pretty much everybody'



5G disappointed consumers and phone carriers. Who's cashing in?

BI Business Insider · Follow

3/14/2025

After astronomic levels of hype, the fifth generation of mobile networks has been a letdown for mai
hasn't even made phone carriers much money. How'd we end up here? And who really benefitted fr

Research Questions

Ground Reality of Today's 5G?

- What is the current state of 5G coverage?
- What speeds do users typically achieve?
- What are the latency characteristics?
- What are the different operator policies?

Can 5G Support “Killer Apps”?

- Evaluate if current 5G networks meet stringent QoE requirements of the ”killer apps”?

How to Enable 5G “Killer Apps”?

- Explore solutions when current 5G falls short in enabling 5G “killer apps"

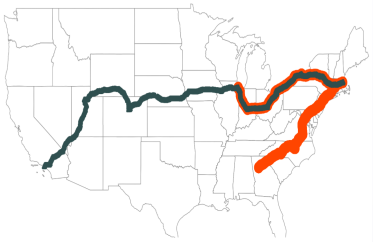
Research Contributions

1a. Large-scale Measurement Campaigns to Study *5G Coverage, Throughput, and Latency*

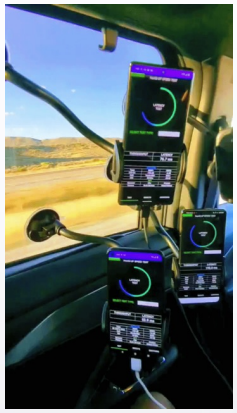
Active Measurements in Mobility Scenarios

Multiple cross-country driving campaigns

[ACM IMC 2023 (25%), ACM IMC 2025 (24%*), ACM CoNEXT 2025 (17%*), PAM 2026 (31%*)]*







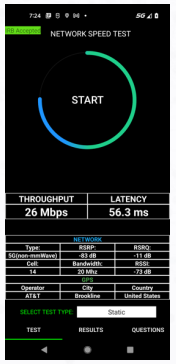
**acceptance rate at conferences*

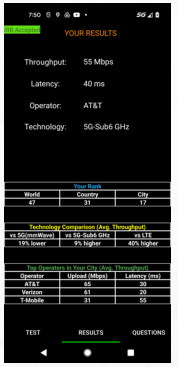


Crowd-Sourced Measurements Using Custom App

US, EU, & Canada institutional collaboration for longitudinal & cross-sectional 5G performance tracking

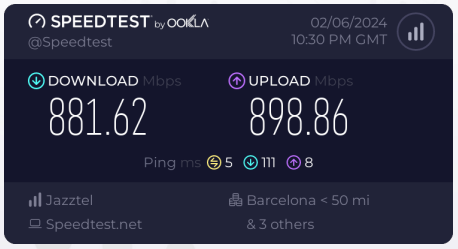
[IEEE/IFIP Networking 2024, Elsevier COMCOM]





Collaboration with Ookla

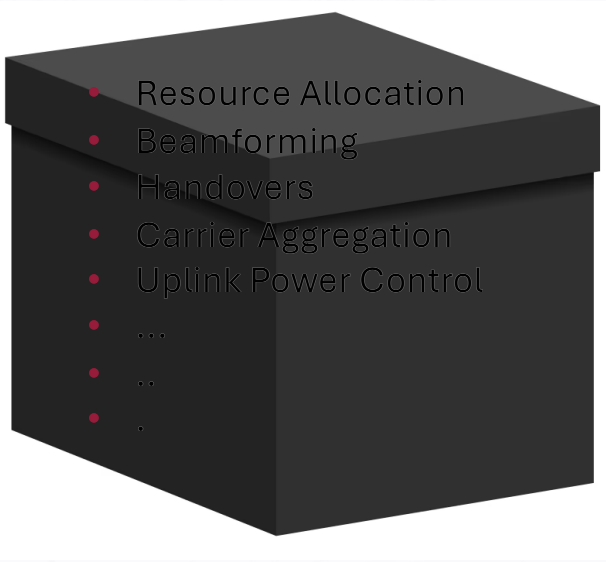
5G evolution study via *Ookla Speedtest* Measurements *[ACM IMC 2025 (24%*)]*



Research Contributions

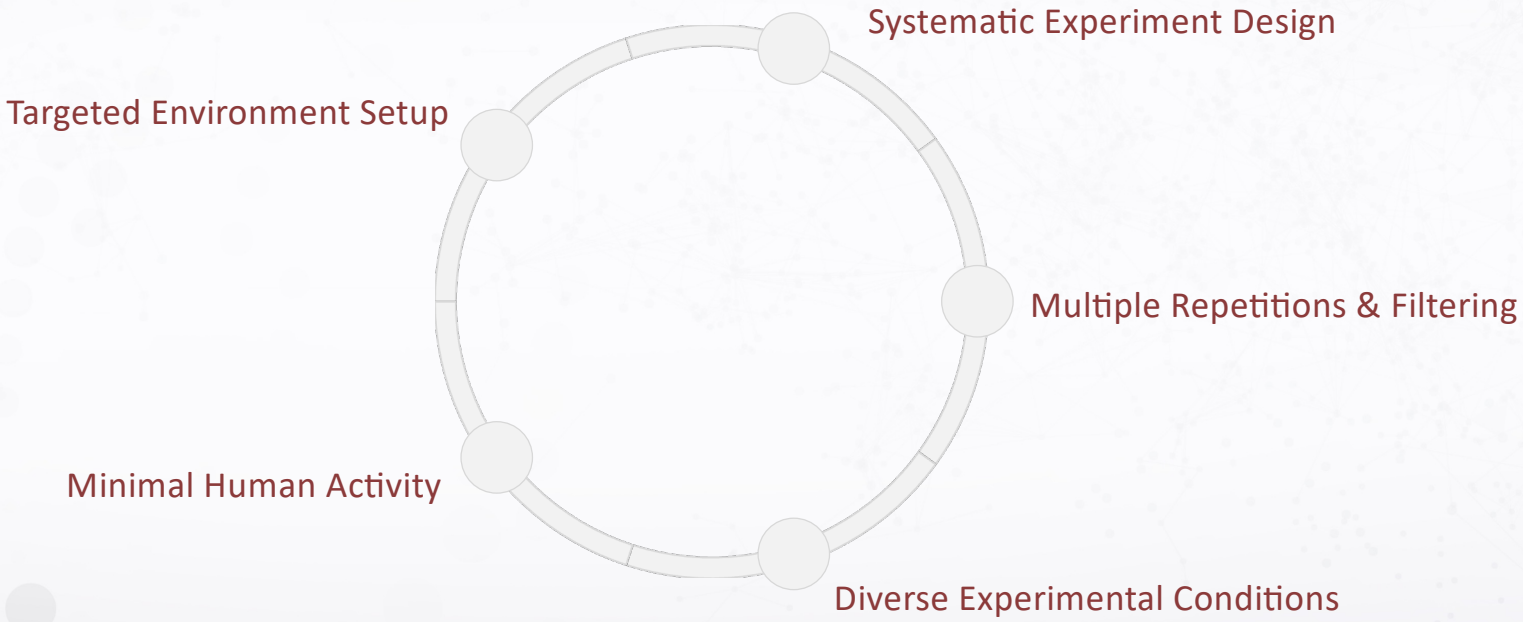
1b. Demystifying Operator Policies





Research Contributions

1b. Controlled Experiments in Multiple Cities to ***Demystify Operator Policies***

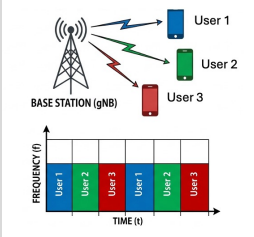


Research Contributions

1b. Controlled Experiments in Multiple Cities to ***Demystify Operator Policies***

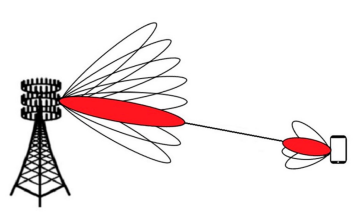
Resource Allocation Policies in 5G mmWave Networks

[IEEE/ACM ToN 2025]



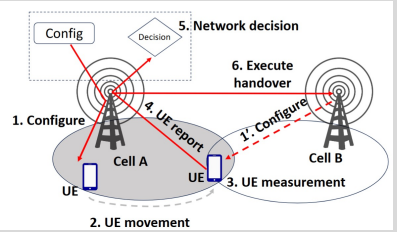
Beamforming Policies in 5G mmWave Networks

[ACM CoNEXT 2025 (17%)]*



Handover Policies in Operational 5G Networks

[Under submission in IEEE ToN]



**acceptance rate at conferences*

Research Contributions

2. Exploring 5G's Capability to Support "5G killer apps"

Evaluating 5G "Killer Apps" in Mobility Scenarios

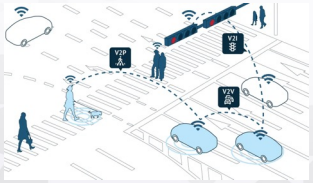
[ACM IMC 2023 (25%), ACM IMC 2025 (24%*)]*



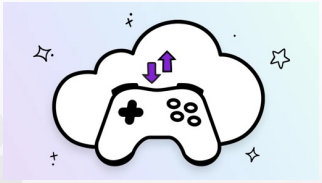
360° Video Streaming



Augmented Reality



Connected Autonomous Vehicles



Cloud Gaming



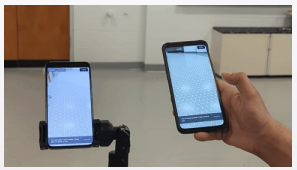
Cloud-Assisted Real-Time Object Detection for AR over 5G mmWave

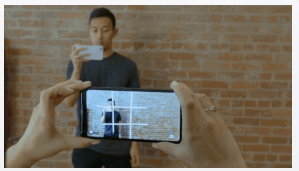
[IEEE MASCOTS (30%)]*



Multi-User AR over 5G

[PAM 2022 (48%), ACM SIGCOMM EMS Workshop]*





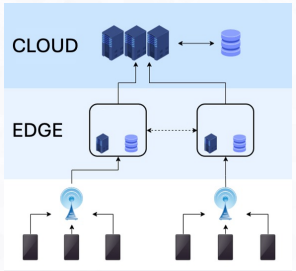
**acceptance rate at conferences*

Research Contributions

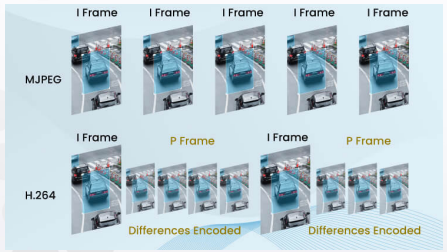
3. Enabling 5G “Killer Apps” over 5G Networks

Optimizations to enhance Application Performance

[ACM IMC 2023 (25%), ACM IMC 2025 (24%*), IEEE MASCOTS (30%*)]*



Using edge servers to reduce end-to-end latency



Application-level optimization techniques to enhance application performance

Starlink as an Alternative

[ACM IMC 2025 (24%), ACM SIGMETRICS (21%*)]*

Head-to-head comparison of Starlink with cellular networks



Rethinking Network Architecture to Enable 5G “Killer Apps”

[IEEE PIMRC 2024, IEEE VTC (Fall) 2025, PAM 2025 (30%)]*

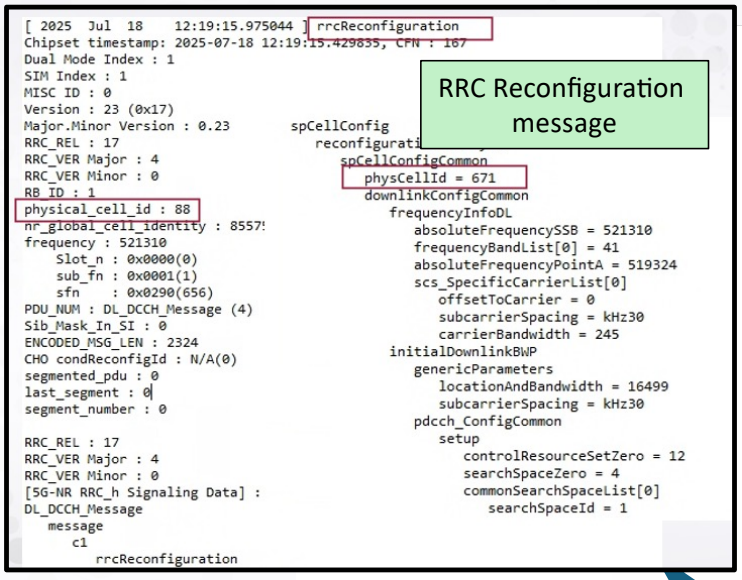
Intelligent scheduling in 5G networks, multi-carrier access to boost user performance

**acceptance rate at conferences*

Research Contributions

- 1a: Large-scale Measurement Campaigns to Study 5G Coverage, Throughput, and Latency
  - Active Measurements in Mobility Scenarios
  - Crowd-Sourced Measurements Using Custom App
  - Collaboration with Ookla
- 1b: Controlled Experiments in Multiple Cities to Demystify Operator Policies
- 2: Exploring 5G's Capability to Support "5G killer apps"
  - Evaluating 5G "Killer Apps" in Mobility Scenarios
  - Cloud-Assisted Real-Time Object Detection for AR over 5G mmWave
  - Multi-User AR over 5G
- 3:Enabling 5G "Killer Apps" over 5G Networks
  - Optimizations to enhance Application Performance
  - Starlink as an Alternative
  - Rethinking Network Architecture to Enable 5G "Killer Apps"

Measurement Testbed



RRC Reconfiguration message

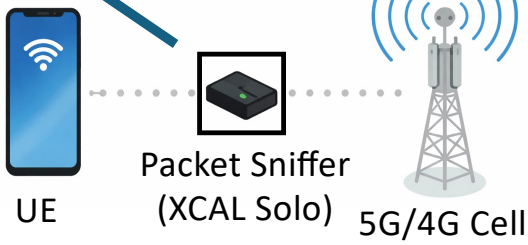
Raw 3GPP Signaling Messages

Parse

TIMESTAMP	Lon	Lat	5G Cell ID	5G RSRP
2025-07-18 12:19:14.000	-106.5836	35.10318	88	-88.18
2025-07-18 12:19:14.500	-106.5835	35.103142	88	-92.03
2025-07-18 12:19:15.000	-106.5835	35.103142	88	-95.32
2025-07-18 12:19:15.500	-106.5835	35.103142	88	-105.76
2025-07-18 12:19:15.946	-106.5832	35.10305		
2025-07-18 12:19:15.975	-106.5832	35.103046		
2025-07-18 12:19:15.975	-106.5832	35.103046		
2025-07-18 12:19:15.975	-106.5832	35.103046		
2025-07-18 12:19:16.000	-106.5832	35.103043	671	-87.08
2025-07-18 12:19:16.044	-106.5831	35.103031		
2025-07-18 12:19:16.044	-106.5831	35.103031		
2025-07-18 12:19:16.044	-106.5831	35.103031		
2025-07-18 12:19:16.044	-106.5831	35.103031		
2025-07-18 12:19:16.045	-106.5831	35.103031		
2025-07-18 12:19:16.045	-106.5831	35.103031		
2025-07-18 12:19:16.500	-106.5829	35.10294	671	-90.16
2025-07-18 12:19:16.806	-106.5828	35.102905		
2025-07-18 12:19:16.806	-106.5828	35.102905		
2025-07-18 12:19:16.806	-106.5828	35.102905		
2025-07-18 12:19:17.000	-106.5827	35.102882	671	-87.15
2025-07-18 12:19:17.500	-106.5826	35.102829	671	-87.54
2025-07-18 12:19:18.000	-106.5824	35.102776	671	-87.31
2025-07-18 12:19:18.500	-106.5823	35.102718	671	-87.81

Handover from cell 88 to 671 with RSRP information

Processed Key Performance Indicators (KPI)



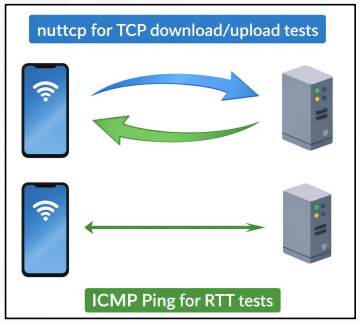
UE

Packet Sniffer (XCAL Solo)

5G/4G Cell

Measurement Testbed

Readily Available Applications



Puffer

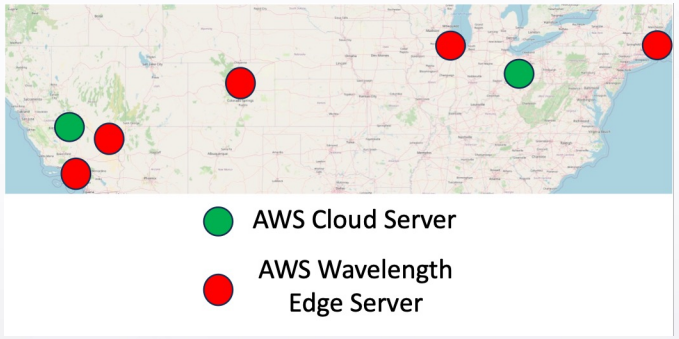
360º video streaming tests

aws

AWS EC2 Servers

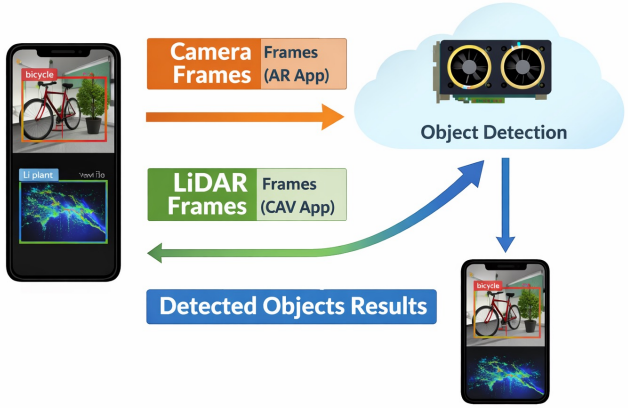


Cloud gaming tests



Custom-built Applications

Custom App 1: Augmented Reality

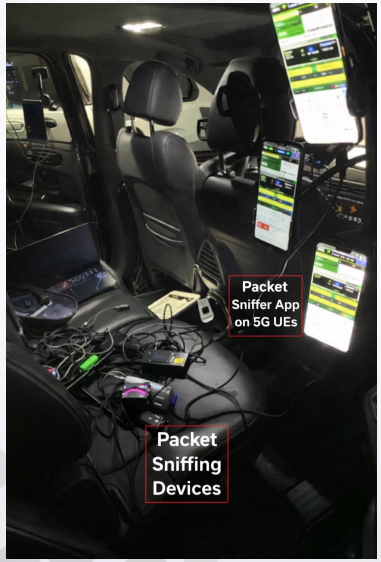


Custom App 2: Connected Autonomous Vehicle

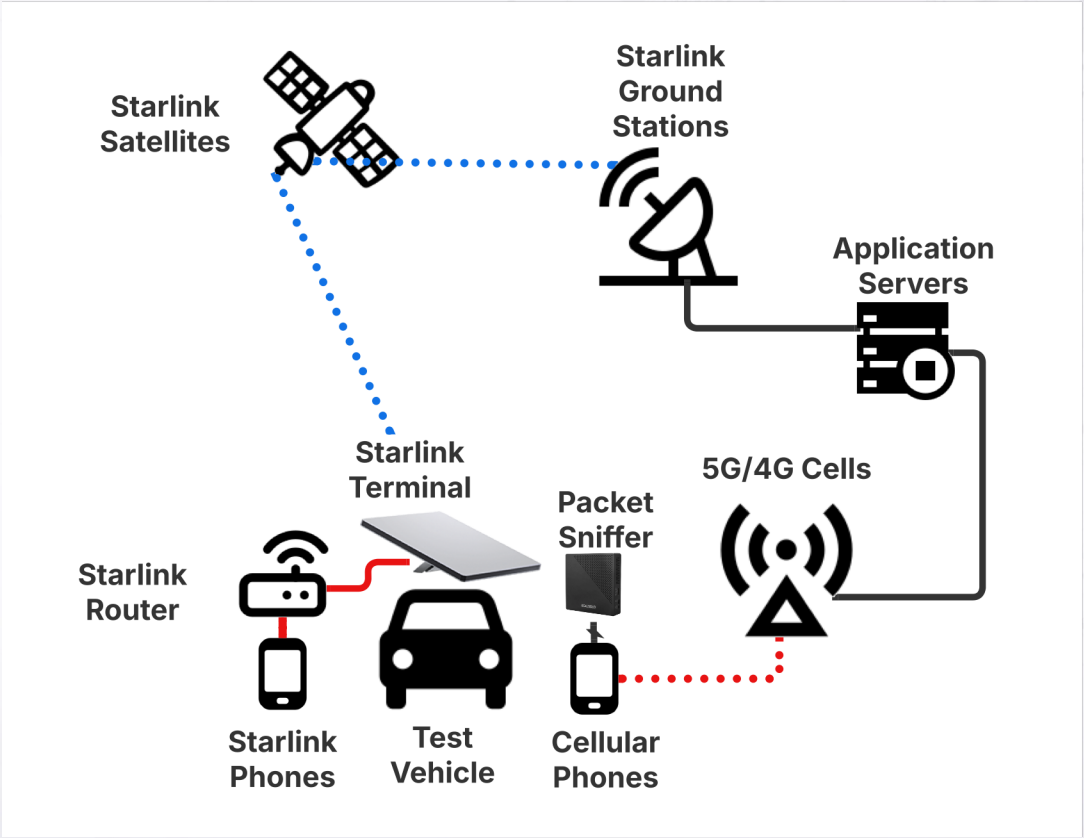
Measurement Testbed



Starlink terminal on top of car



Testbed inside the car



5G Measurement Campaigns

- 4 cross-country driving trips
  - Boston to Chicago (2023)
  - Boston to Atlanta (2023)
  - Los Angeles to Boston (2022, 2024)



- 3 multi-day trips in rural areas (2024)
  - Alaska, Maui HI, New England



- Smaller measurement campaigns in select cities (2021-2024)
  - Boston, Atlanta, Charlotte, Chicago, Indianapolis, Miami, Las Vegas



Performance of Cellular Networks on the Wheels*#



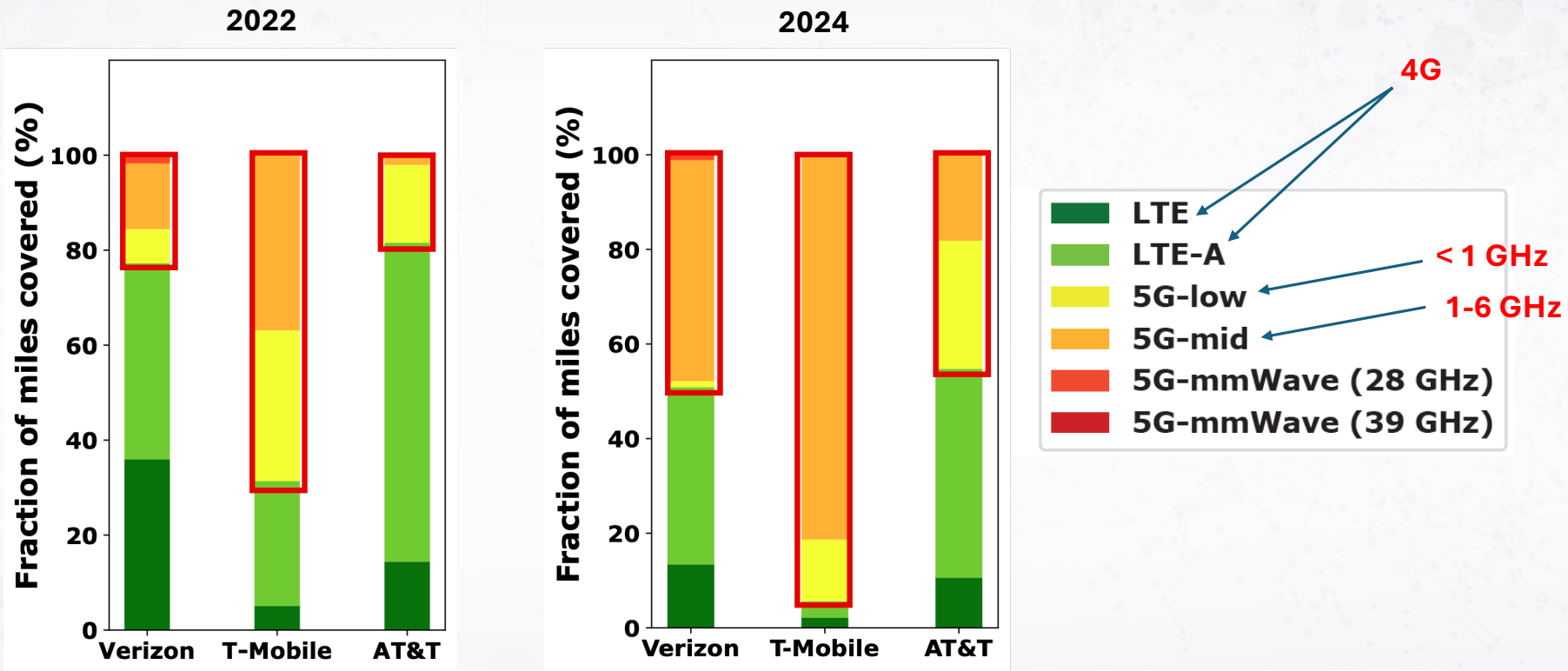
- **First** in-depth study of mobile network performance and coverage of all **3 major US operators** while driving cross-continental US
- Comprehensive evolution study of cellular coverage and performance across 2 years

Moinak Ghoshal, Imran Khan, Z. Jonny Kong, Phuc Dinh, Jiayi Meng, Y. Charlie Hu, and Dimitrios Koutsonikolas. Performance of Cellular Networks on the Wheels. In Proc. of ACM IMC 2023.

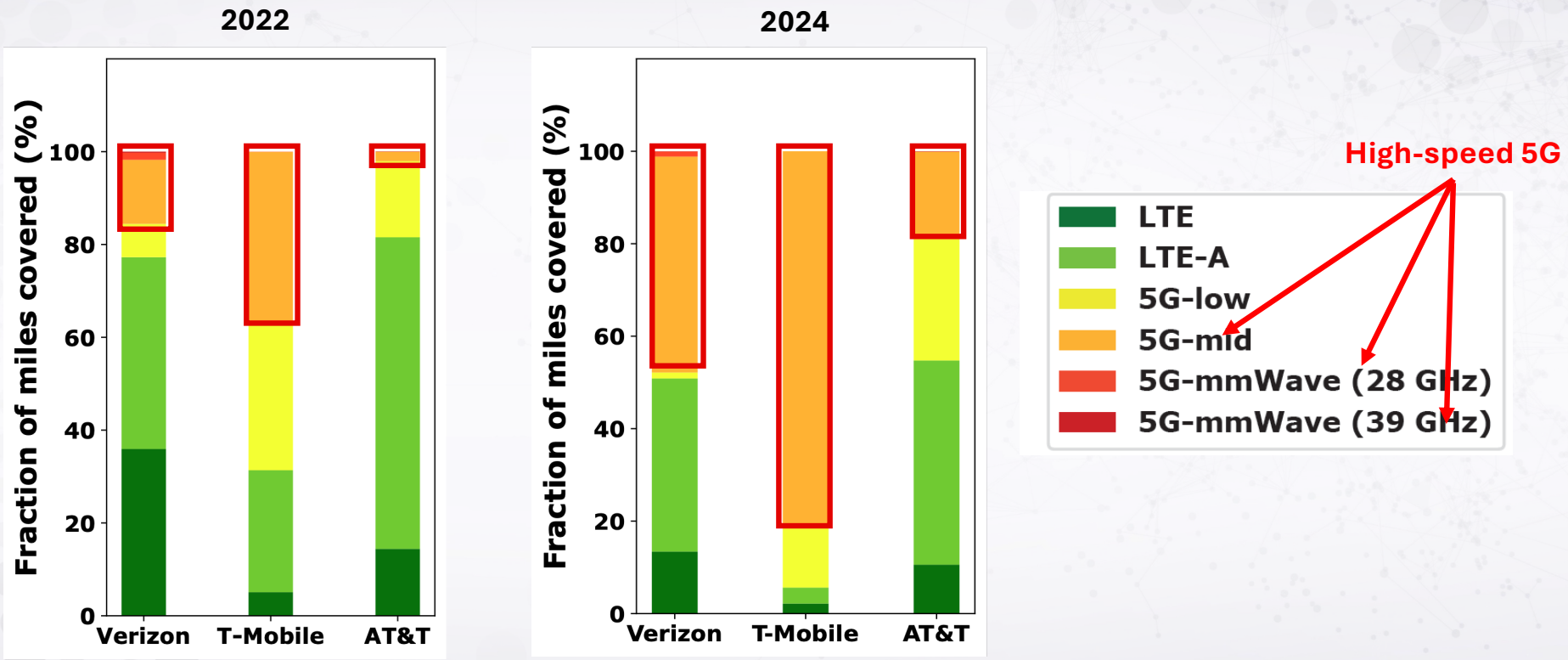
#Moinak Ghoshal, Omar Basit, Imran Khan, Z. Jonny Kong, Sizhe Wang, Yufei Feng, Phuc Dinh, Y. Charlie Hu, and Dimitrios Koutsonikolas. Replication: Performance of Cellular Networks on the Wheels. In Proc. of ACM IMC 2025.

Network Coverage

Overall Coverage



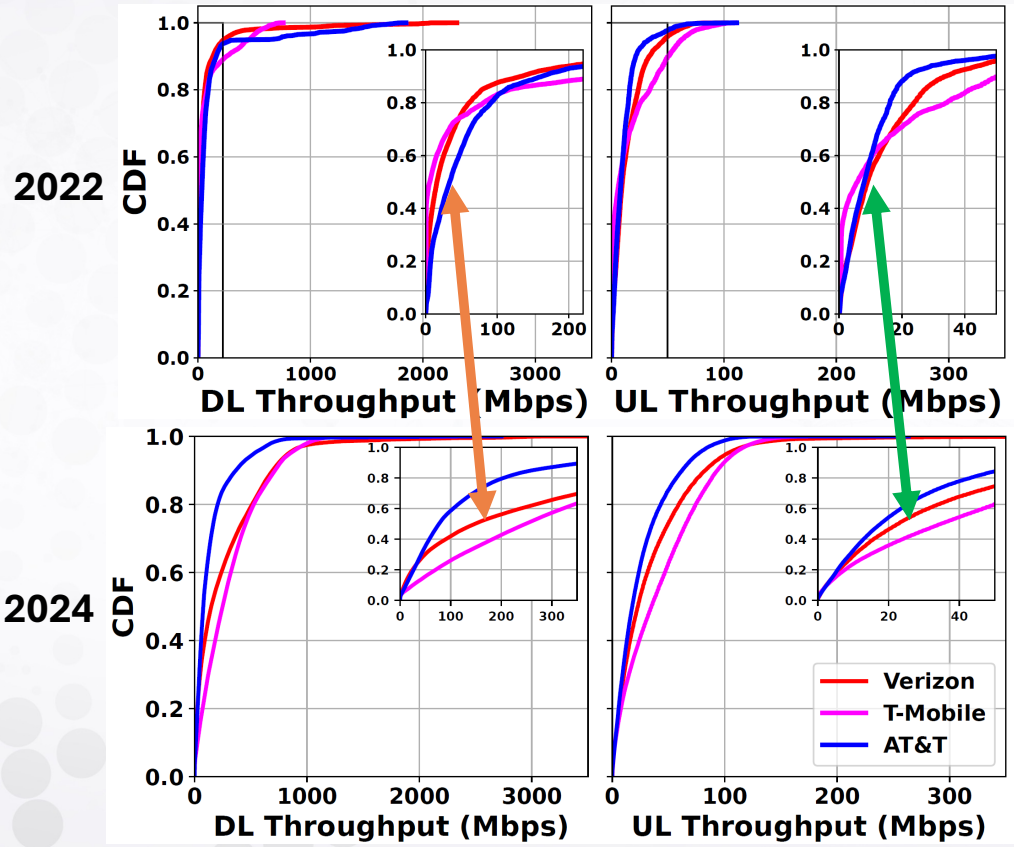
Overall Coverage



- High-speed 5G coverage increased significantly from 2022 to 2024

Network Performance

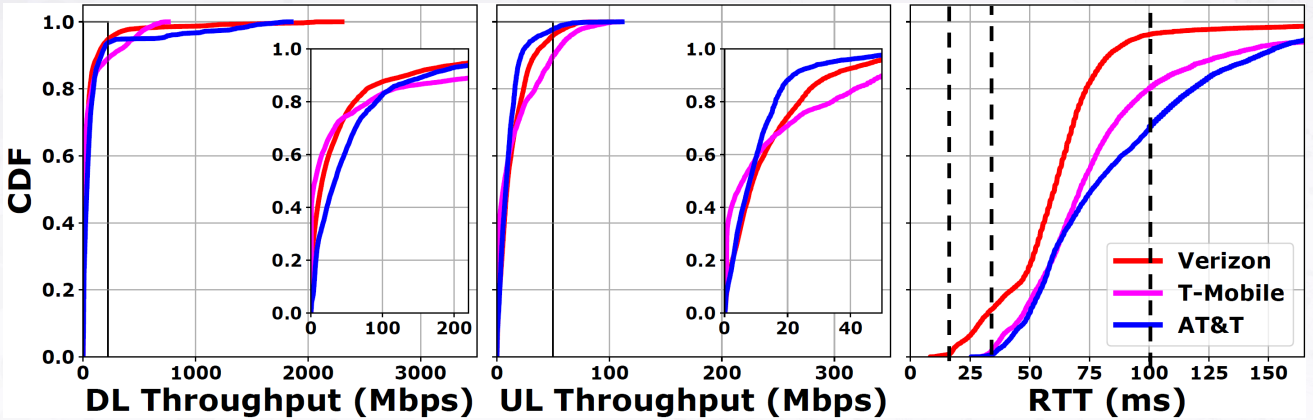
Overall Performance



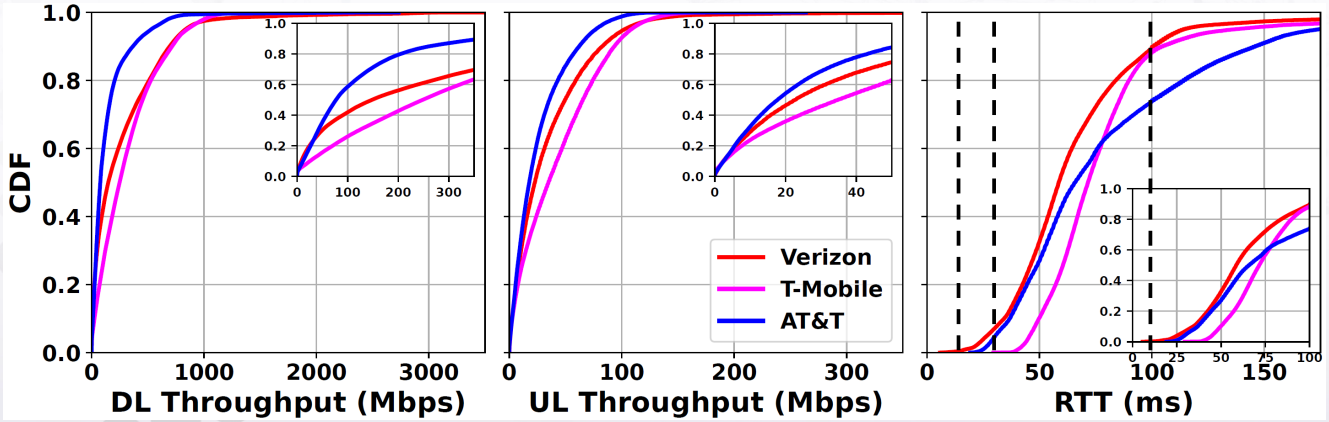
- Multi-fold increase in throughput from 2022 to 2024
  - Median DL throughput improvement → 2x to 40x
  - Median UL throughput improvement → 2x to 6x

Overall Performance

2022



2024



Round Trip Time (RTT) remains largely the same between 2022 and 2024

- CAV E2E latency < 100 ms
- AR @30 FPS → E2E latency 33 ms
- Motion to photon latency < 15 ms

Research Contributions

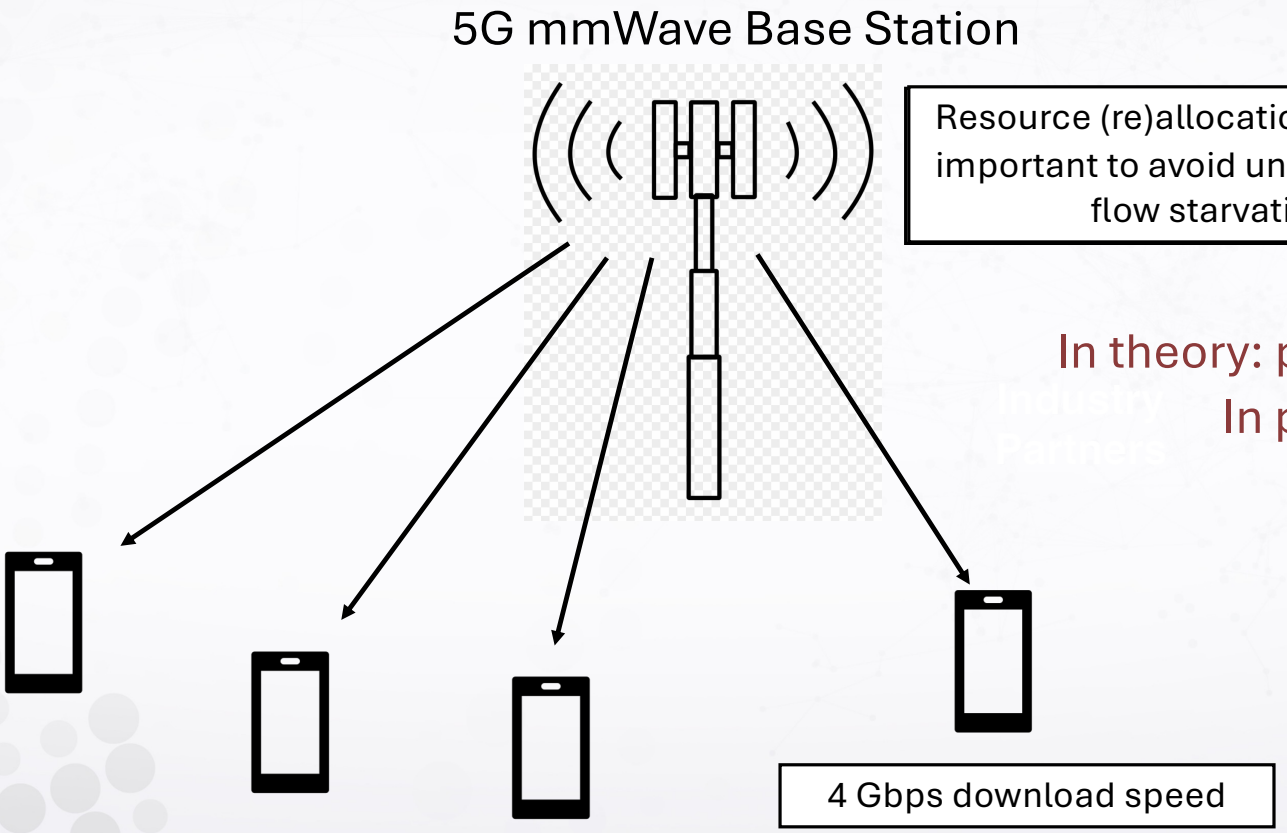
- 1a: Large-scale Measurement Campaigns to Study 5G Coverage, Throughput, and Latency
  - Active Measurements in Mobility Scenarios ✓
  - Crowd-Sourced Measurements Using Custom App
  - Collaboration with Ookla

- 1b: Controlled Experiments in Multiple Cities to Demystify Operator Policies

- 2: Exploring 5G's Capability to Support "5G killer apps"
  - Evaluating 5G "Killer Apps" in Mobility Scenarios
  - Cloud-Assisted Real-Time Object Detection for AR over 5G mmWave
  - Multi-User AR over 5G

- 3:Enabling 5G "Killer Apps" over 5G Networks
  - Optimizations to enhance Application Performance
  - Starlink as an Alternative
  - Rethinking Network Architecture to Enable 5G "Killer Apps"

Resource Sharing in 5G mmWave Networks



Resource Sharing in 5G mmWave Networks

First systematic study of resource allocation policies in operational 5G mmWave networks[1]



verizon

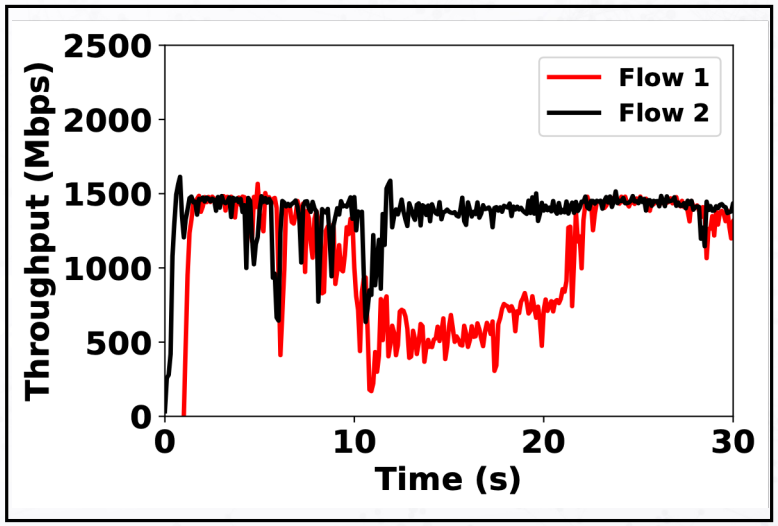
AT&T

[1] *M. Ghoshal, P. Dinh, Y. Han, Y. Feng, D. Koutsonikolas and J. Widmer, "Demystifying Resource Allocation Policies in Operational 5G mmWave Networks," in IEEE Transactions on Networking, vol. 33, no. 2, pp. 745-760, April 2025.*

Motivating Example

One user walks towards & away from BS (Flow 1); One user is static facing BS (Flow 2)

- Observations
  - Total capacity: **3 Gbps**
  - Each flow allocated fixed capacity: **1.5 Gbps**
  - One flow drops due to mobility: other flow still **<= 1.5 Gbps**

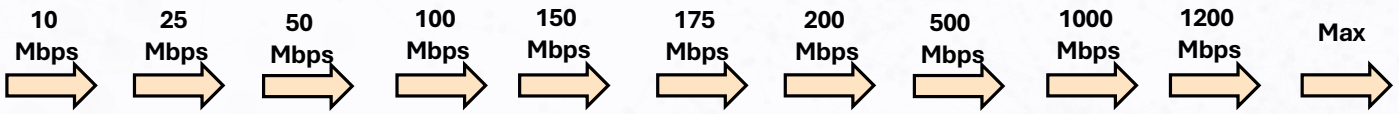


- Conclusions
  - Two flows do not compete in a shared queue
  - Operator neither employs proportional fairness nor tries to maximize total capacity
  - Resource allocation governed strictly by simple threshold-based policies

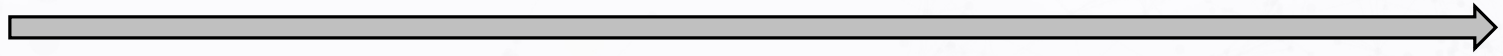
What are these policies?

Policies Across Operators and Cities

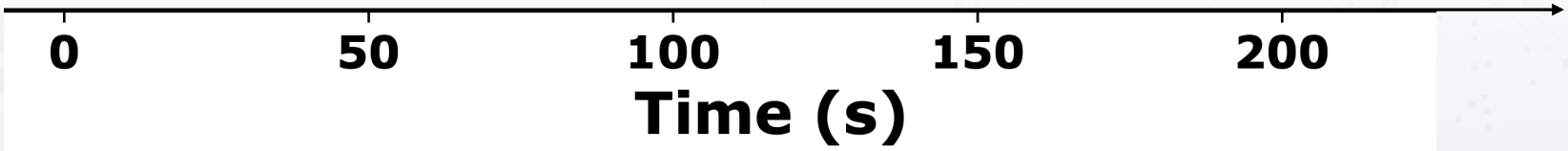
Objective: Observe how the network reacts in presence of **heterogenous traffic demands**



Flow 2: Intermittent varying traffic →10 to 220 s

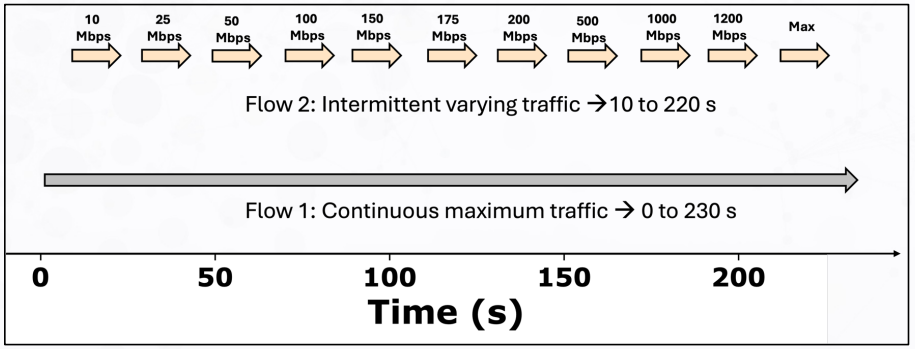


Flow 1: Continuous maximum traffic → 0 to 230 s

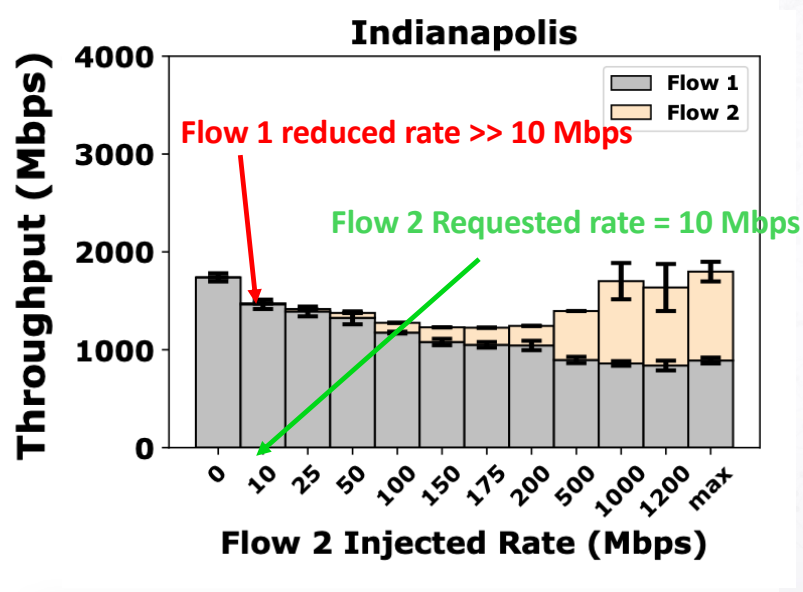


Policies Across Operators and Cities

Objective: Observe how the network reacts in presence of **heterogenous traffic demands**



verizon



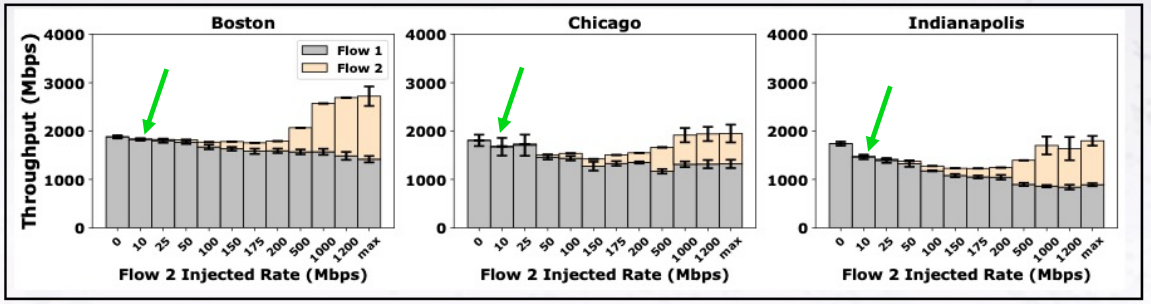
Policy: Overprovisioning of resources

Policies Across Operators and Cities

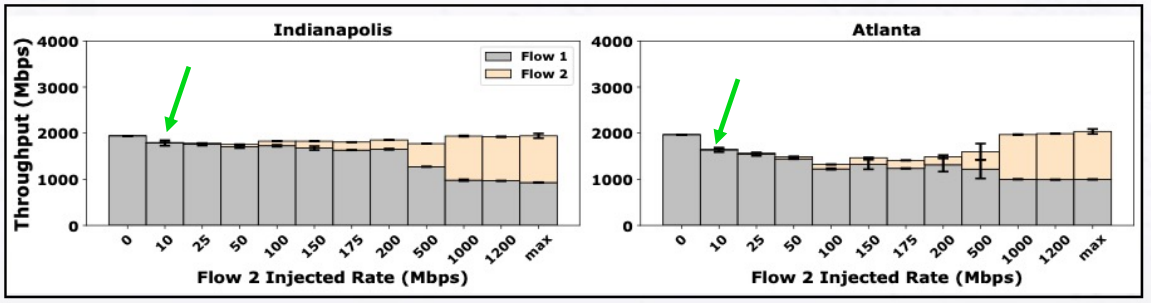
Objective: Observe how the network reacts in presence of **heterogenous traffic demands**

Common Policy: Overprovisioning of resources

Verizon



AT&T



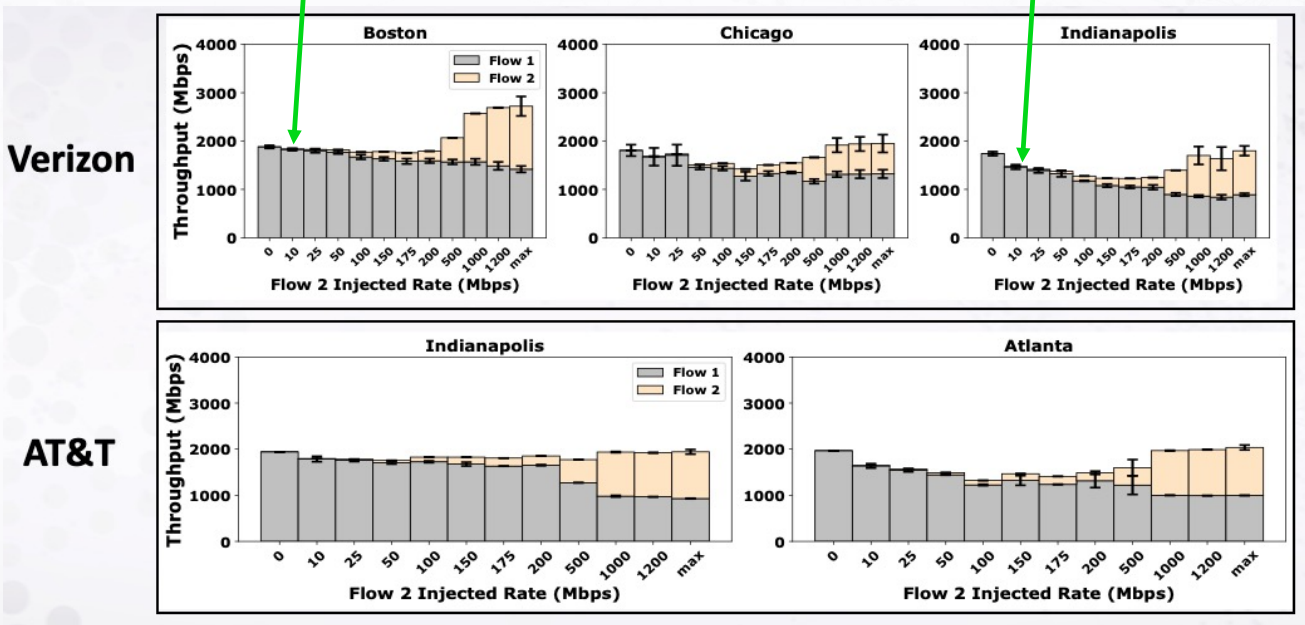
Policies Across Operators and Cities

Objective: Observe how the network reacts in presence of **heterogenous traffic demands**

Different Policy: Different safety margin for overprovisioning

Flow 1 reduced rate ~ 10s of Mbps

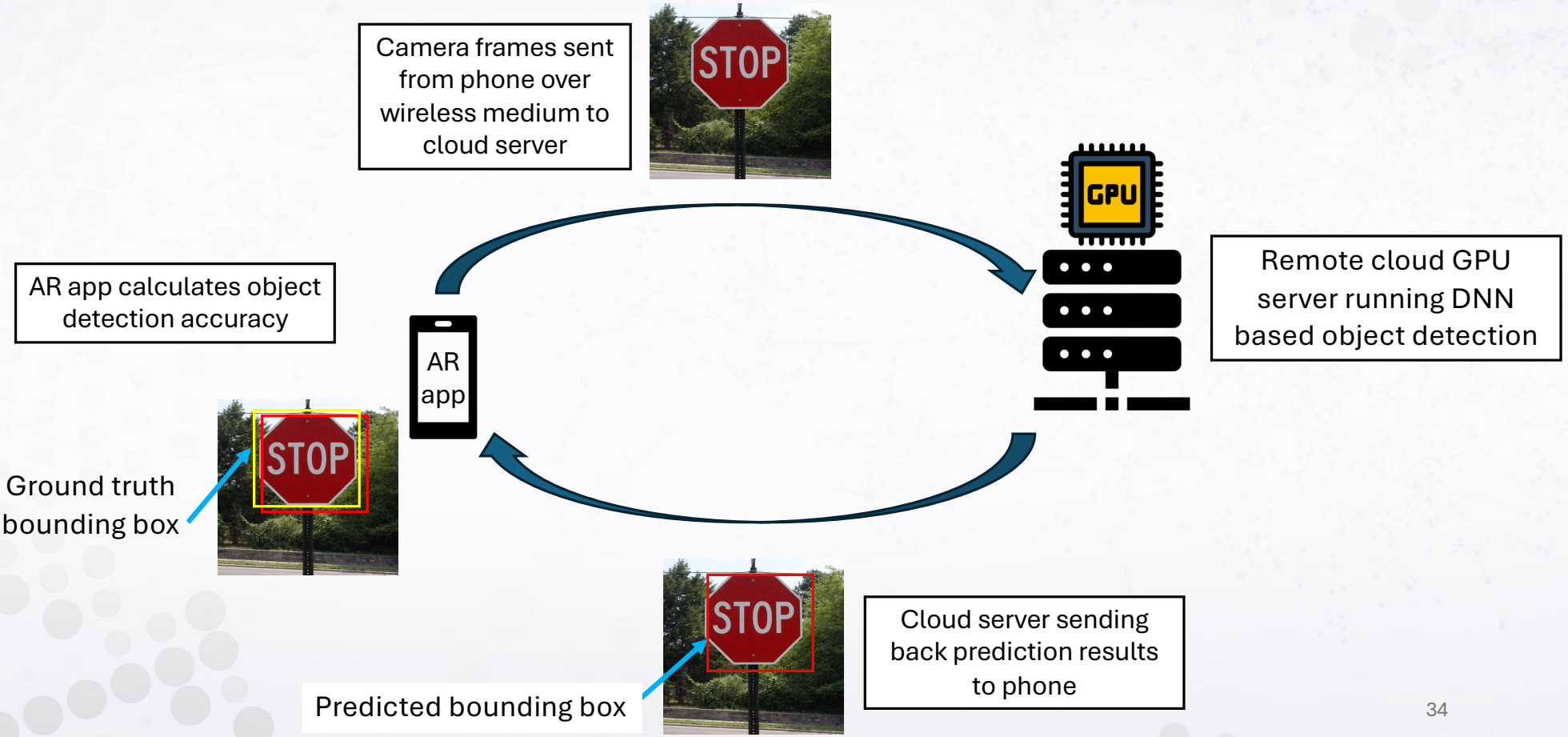
Flow 1 reduced rate ~ 100s of Mbps



Research Contributions

- 1a: Large-scale Measurement Campaigns to Study 5G Coverage, Throughput, and Latency
  - Active Measurements in Mobility Scenarios ✓
  - Crowd-Sourced Measurements Using Custom App
  - Collaboration with Ookla
- 1b: Controlled Experiments in Multiple Cities to Demystify Operator Policies ✓
- 2: Exploring 5G's Capability to Support "5G killer apps"
  - Evaluating 5G "Killer Apps" in Mobility Scenarios
  - Cloud-Assisted Real-Time Object Detection for AR over 5G mmWave
  - Multi-User AR over 5G
- 3:Enabling 5G "Killer Apps" over 5G Networks
  - Optimizations to enhance Application Performance
  - Starlink as an Alternative
  - Rethinking Network Architecture to Enable 5G "Killer Apps"

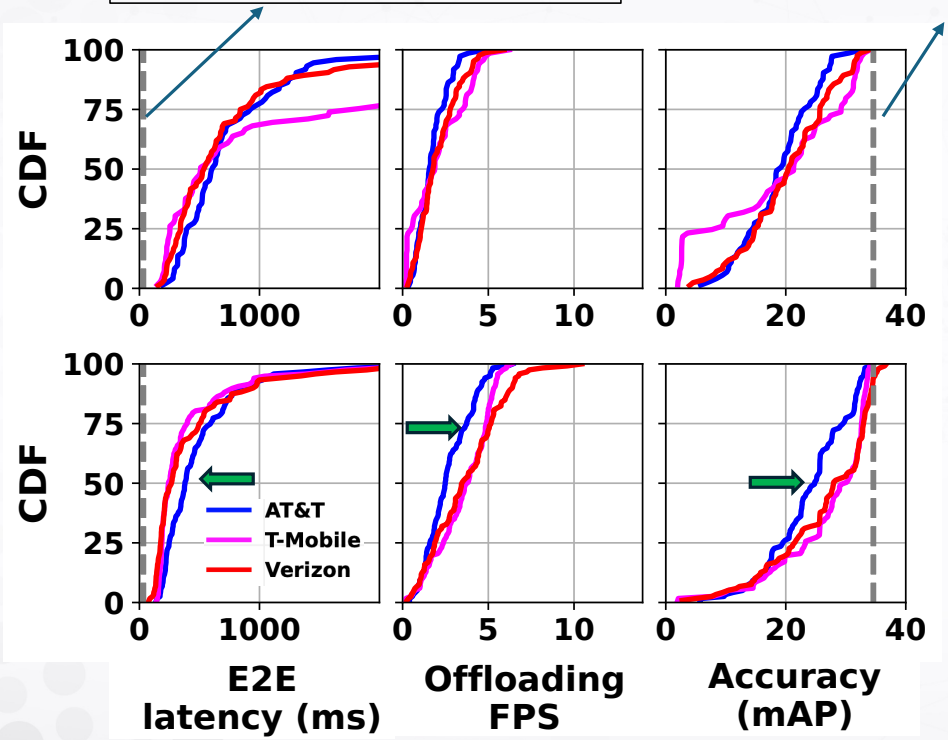
Augmented Reality (Uplink Oriented)



Augmented Reality Performance

30 FPS → 33 ms E2E latency

90% of offline accuracy



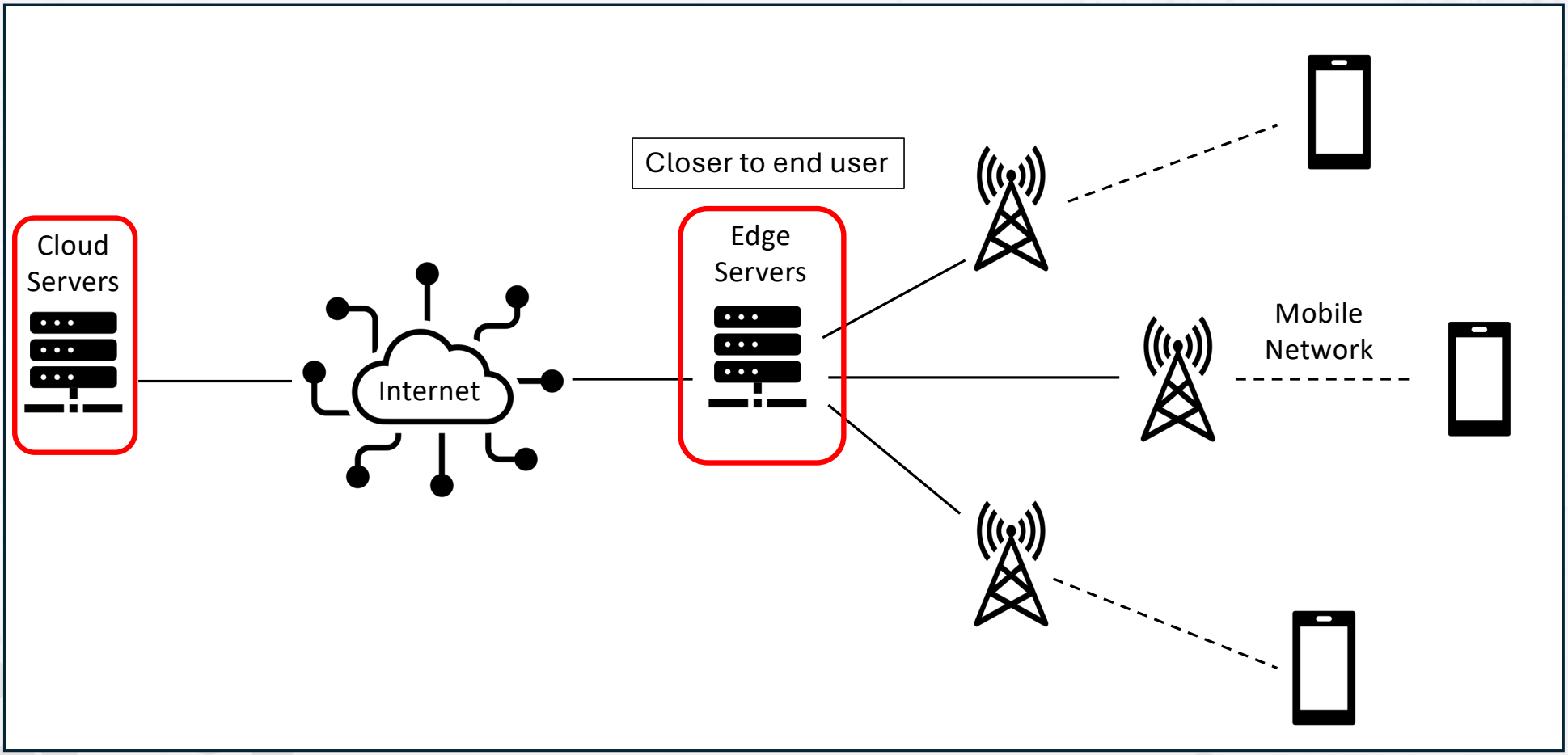
- Substantial increase in AR performance in 2024 across all metrics
  - **Accuracy still far from satisfactory!**

Out of the box AR app does not work over today's cellular networks

Research Contributions

- 1a: Large-scale Measurement Campaigns to Study 5G Coverage, Throughput, and Latency
  - Active Measurements in Mobility Scenarios ✓
  - Crowd-Sourced Measurements Using Custom App
  - Collaboration with Ookla
- 1b: Controlled Experiments in Multiple Cities to Demystify Operator Policies ✓
- 2: Exploring 5G's Capability to Support "5G killer apps"
  - Evaluating 5G "Killer Apps" in Mobility Scenarios ✓
  - Cloud-Assisted Real-Time Object Detection for AR over 5G mmWave
  - Multi-User AR over 5G
- 3:Enabling 5G "Killer Apps" over 5G Networks
  - Optimizations to enhance Application Performance
  - Starlink as an Alternative
  - Rethinking Network Architecture to Enable 5G "Killer Apps"

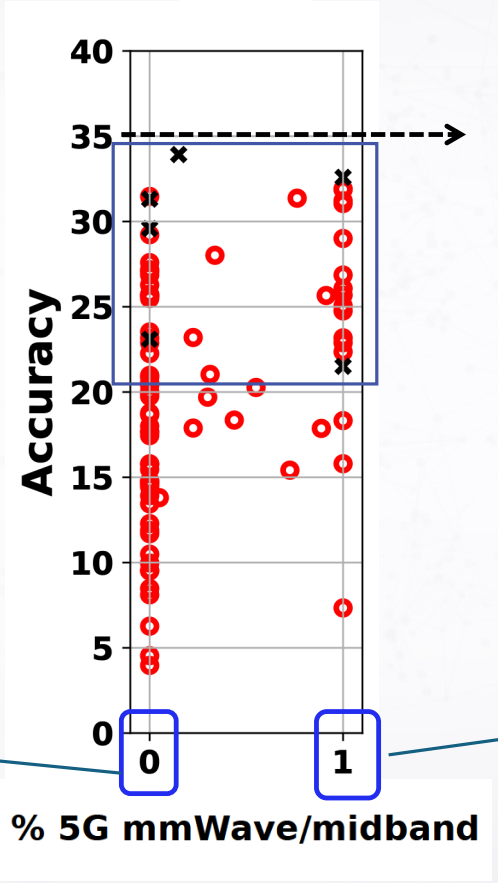
Impact of Edge Servers



Impact of Edge Servers

Cloud server used

Edge server used



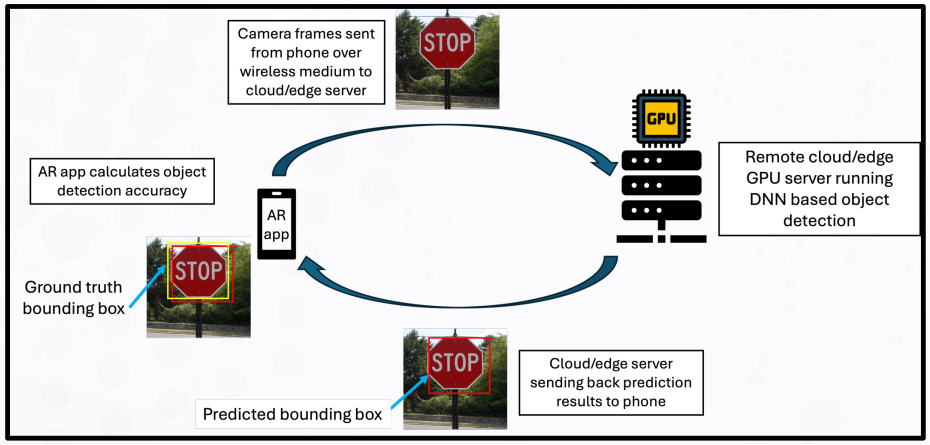
Edge servers help regardless of cellular technology

Still not enough to meet accuracy demands

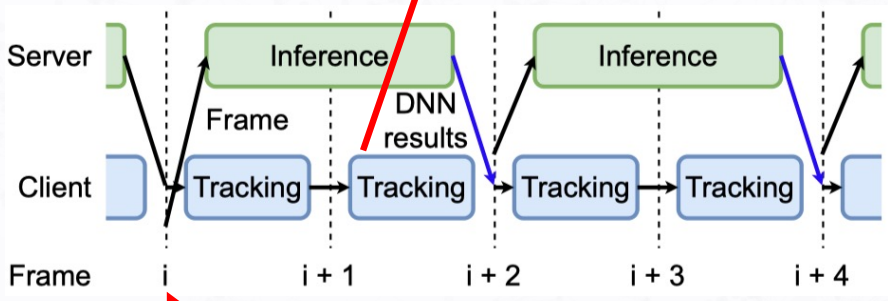
Entire test uses LTE/LTE-A/5G low

Entire test uses high-speed 5G

App-level Optimizations for AR App



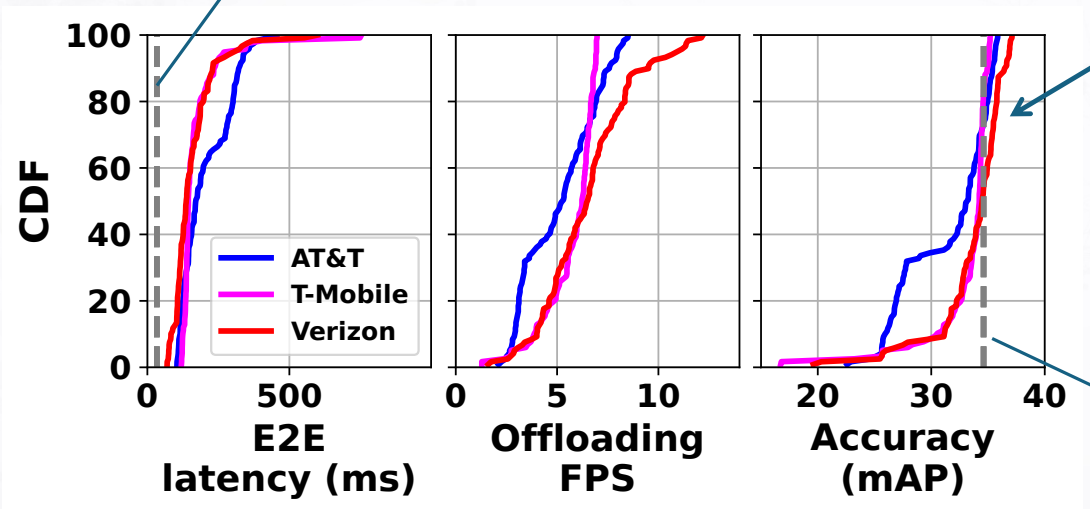
Local tracking:
Updates stale bounding boxes using motion between frames



Compress the frames and send

AR Performance with App-Level Optimizations

30 FPS → 33 ms E2E latency



- App achieves satisfactory performance 25-50% of the time
  - Despite large E2E latency and low frame rate

90% of offline accuracy

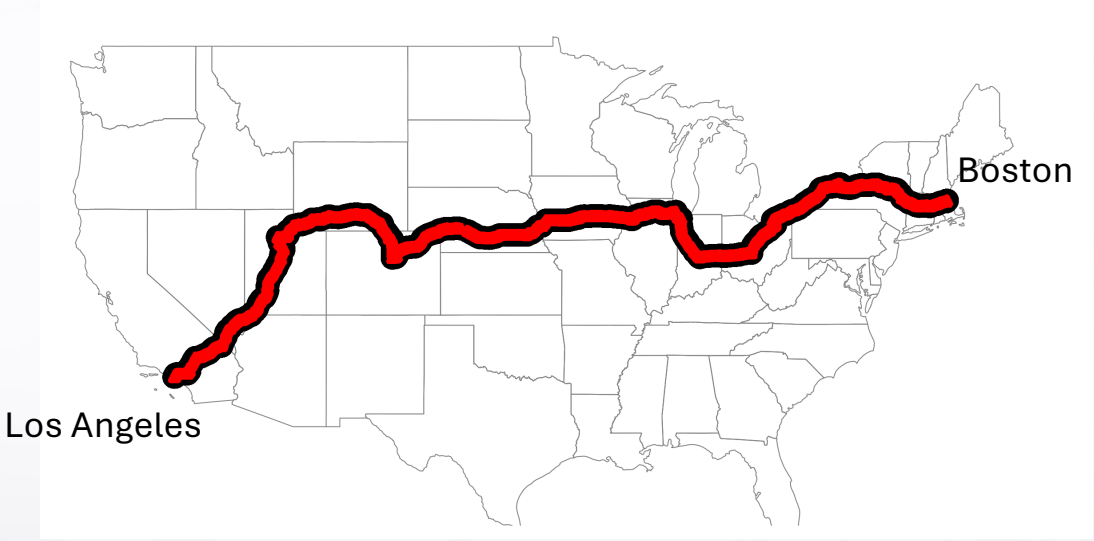
Network improvements + Infrastructural changes + App-level optimizations needed to improve 5G "killer apps" performance

Research Contributions

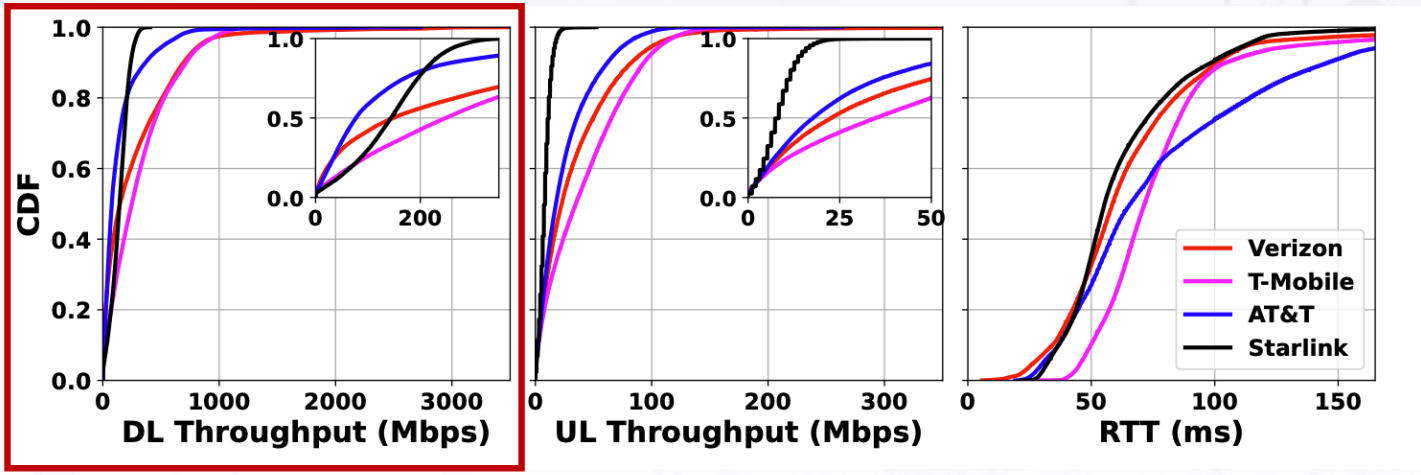
- 1a: Large-scale Measurement Campaigns to Study 5G Coverage, Throughput, and Latency
  - Active Measurements in Mobility Scenarios ✓
  - Crowd-Sourced Measurements Using Custom App
  - Collaboration with Ookla
- 1b: Controlled Experiments in Multiple Cities to Demystify Operator Policies ✓
- 2: Exploring 5G's Capability to Support "5G killer apps"
  - Evaluating 5G "Killer Apps" in Mobility Scenarios ✓
  - Cloud-Assisted Real-Time Object Detection for AR over 5G mmWave
  - Multi-User AR over 5G
- 3:Enabling 5G "Killer Apps" over 5G Networks
  - Optimizations to enhance Application Performance ✓
  - Starlink as an Alternative
  - Rethinking Network Architecture to Enable 5G "Killer Apps"

Starlink vs. Cellular: Network Availability

- Cellular networks: No outage along the route
- Starlink: Outage for approximately 2.03% of the route

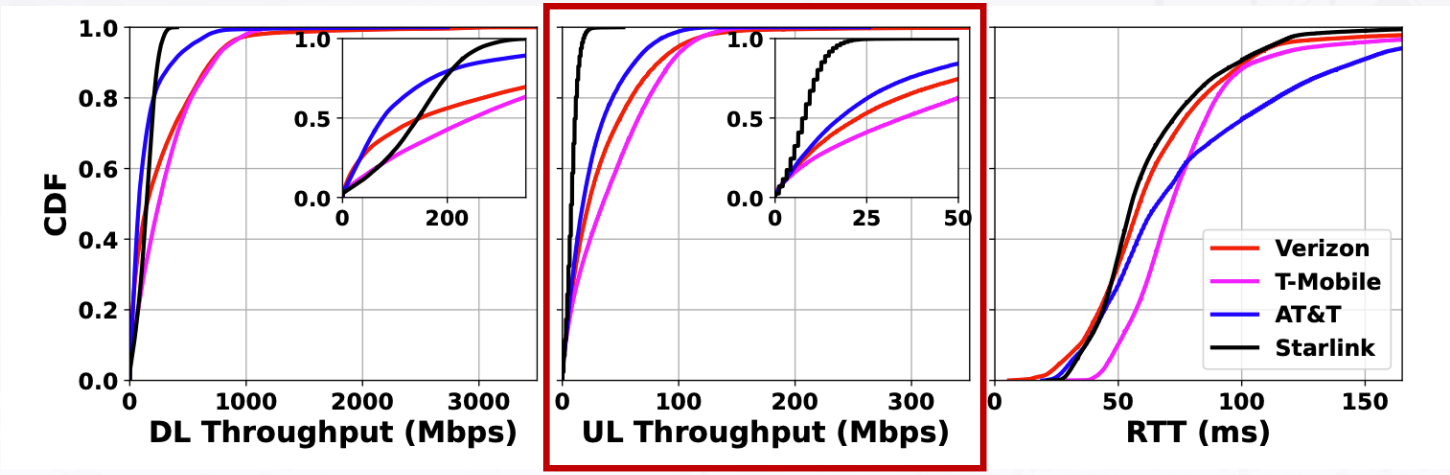


Starlink vs. Cellular: Network Performance



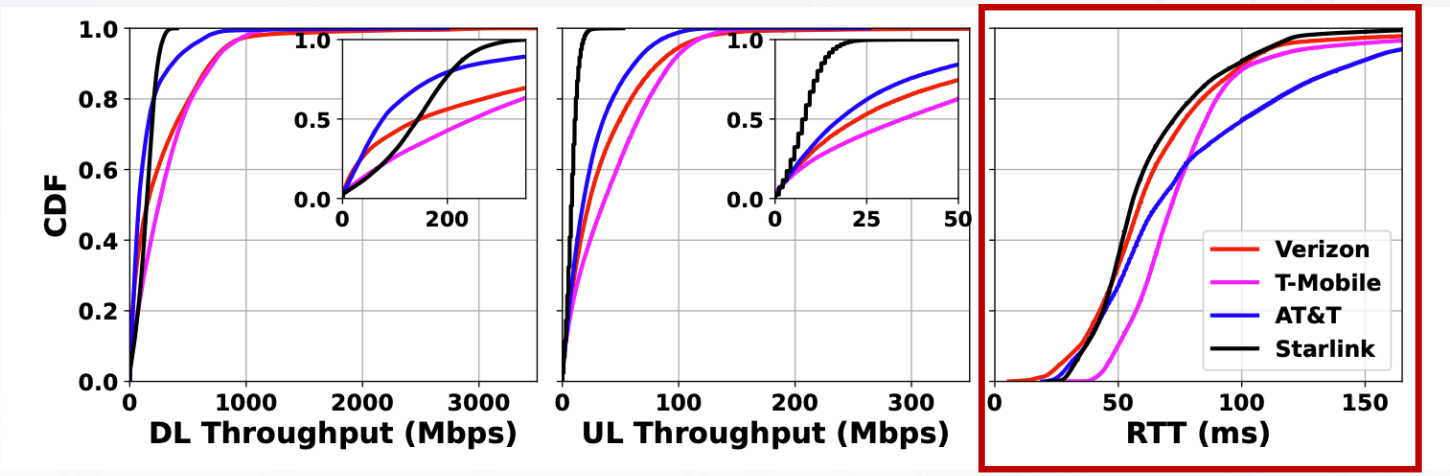
- Downlink Throughput
  - Starlink outperforms AT&T/Verizon/T-Mobile 80%/50%/25% of the time
  - Peak Starlink throughput is 406 Mbps; Peak cellular throughput 2-4 Gbps

Starlink vs. Cellular: Network Performance



- Downlink Throughput
  - Starlink outperforms AT&T/Verizon/T-Mobile 80%/50%/25% of the time
  - Peak Starlink throughput is 406 Mbps; Peak cellular throughput 2-4 Gbps
- Uplink Throughput
  - Starlink achieves the lowest performance among the four networks

Starlink vs. Cellular: Network Performance



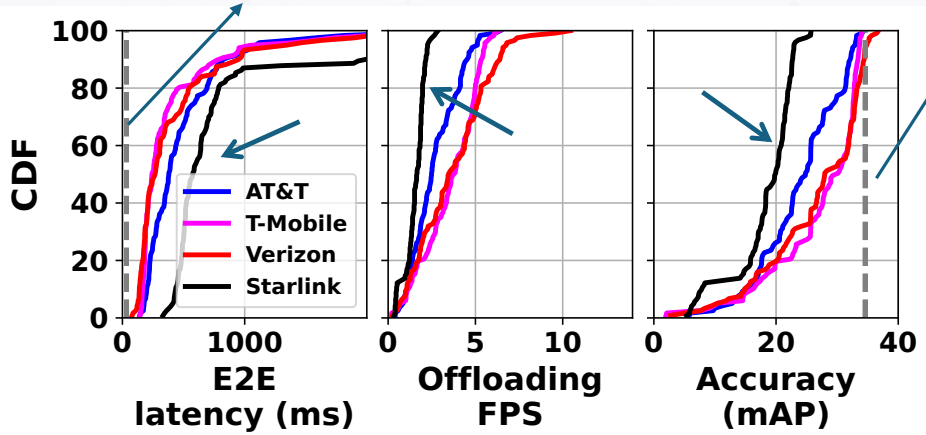
- Downlink Throughput
  - Starlink outperforms AT&T/Verizon/T-Mobile 80%/50%/25% of the time
  - Peak Starlink throughput is 406 Mbps; Peak cellular throughput 2-4 Gbps
- Uplink Throughput
  - Starlink achieves the lowest performance among the four networks
- Starlink RTT **lower** than cellular RTT

Starlink vs. Cellular: AR Performance

30 FPS → 33 ms E2E latency

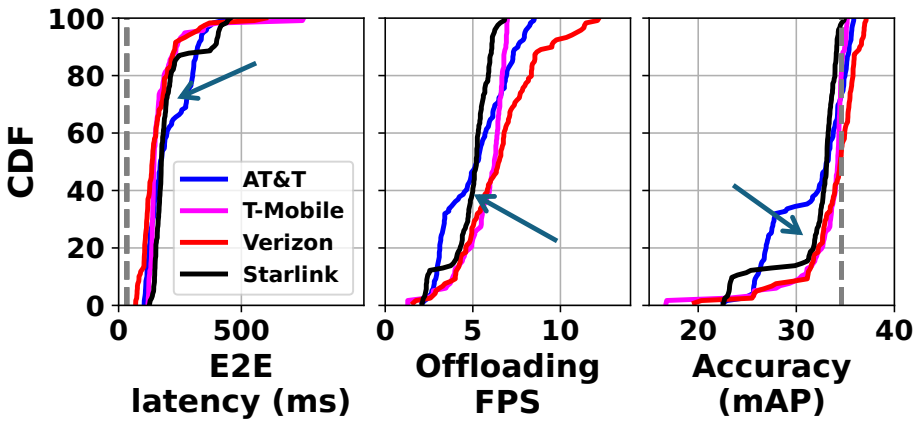
90% of offline accuracy

Raw Camera Frames



- Raw AR frames: Starlink delivers lowest performance
  - Worst uplink throughput
  - Best RTT

Compressed Camera Frames



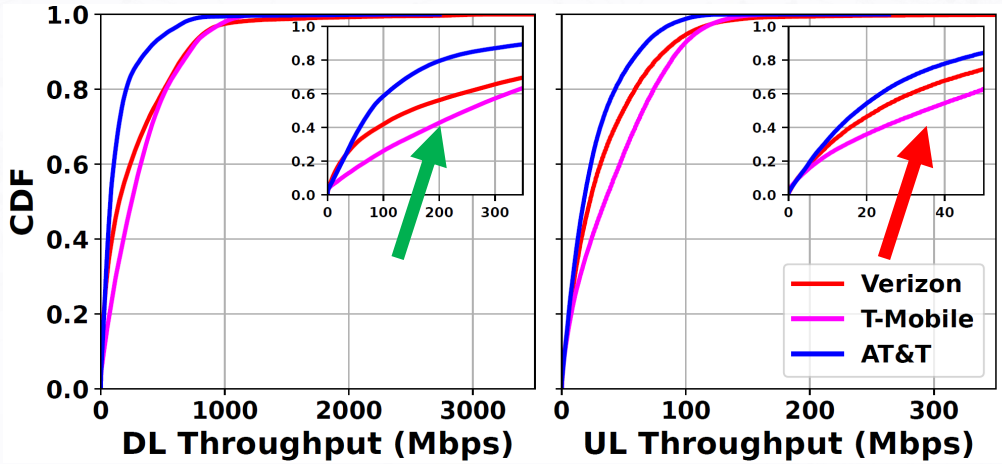
- Compressed AR frames: Starlink performance improves

Research Contributions

- 1a: Large-scale Measurement Campaigns to Study 5G Coverage, Throughput, and Latency
  - Active Measurements in Mobility Scenarios ✓
  - Crowd-Sourced Measurements Using Custom App
  - Collaboration with Ookla
- 1b: Controlled Experiments in Multiple Cities to Demystify Operator Policies ✓
- 2: Exploring 5G's Capability to Support "5G killer apps"
  - Evaluating 5G "Killer Apps" in Mobility Scenarios ✓
  - Cloud-Assisted Real-Time Object Detection for AR over 5G mmWave
  - Multi-User AR over 5G
- 3:Enabling 5G "Killer Apps" over 5G Networks
  - Optimizations to enhance Application Performance ✓
  - Starlink as an Alternative ✓
  - Rethinking Network Architecture to Enable 5G "Killer Apps"

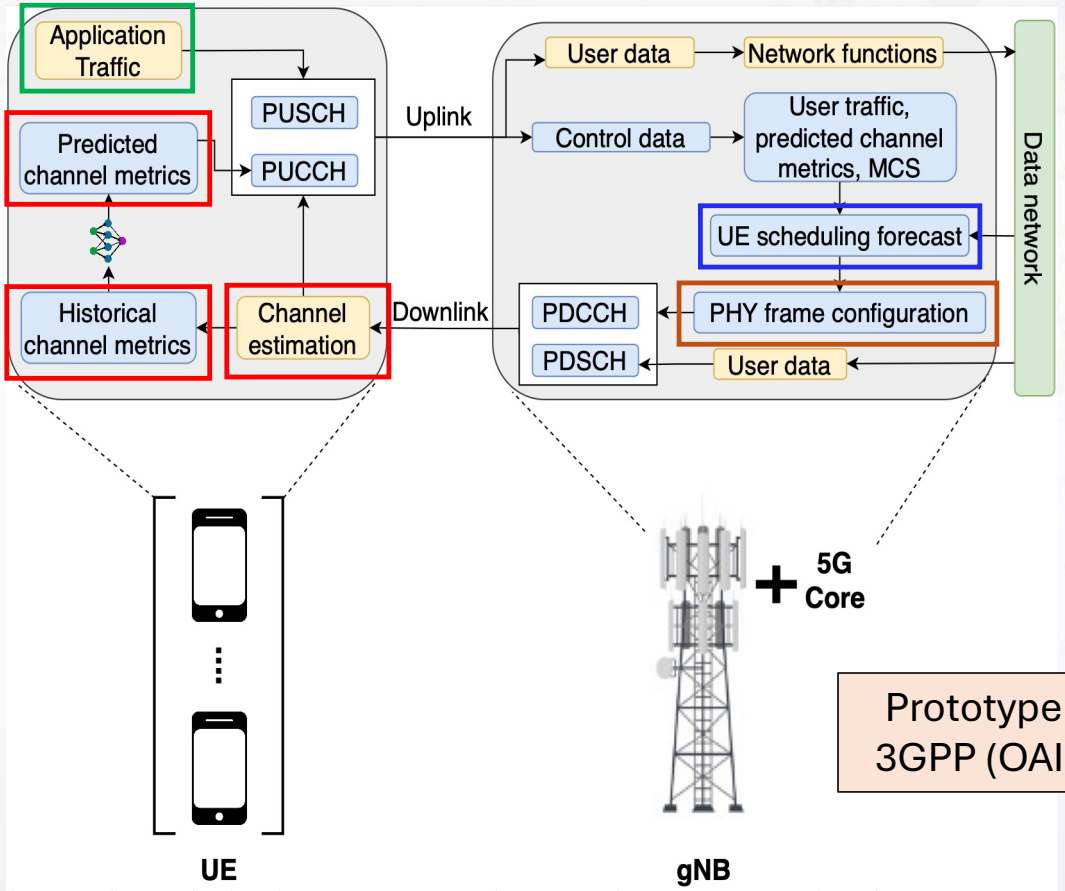
Intelligent Scheduling in 5G Networks

- Problem 1: Asymmetric 5G DL/UL performance



  - But many NextG apps are uplink-oriented (AR, CAVs)
- Problem 2: Operators used static resource allocation configurations
  - But traffic demands vary over time

UE-assisted Proactive Frame Configuration



Existing 5G modules

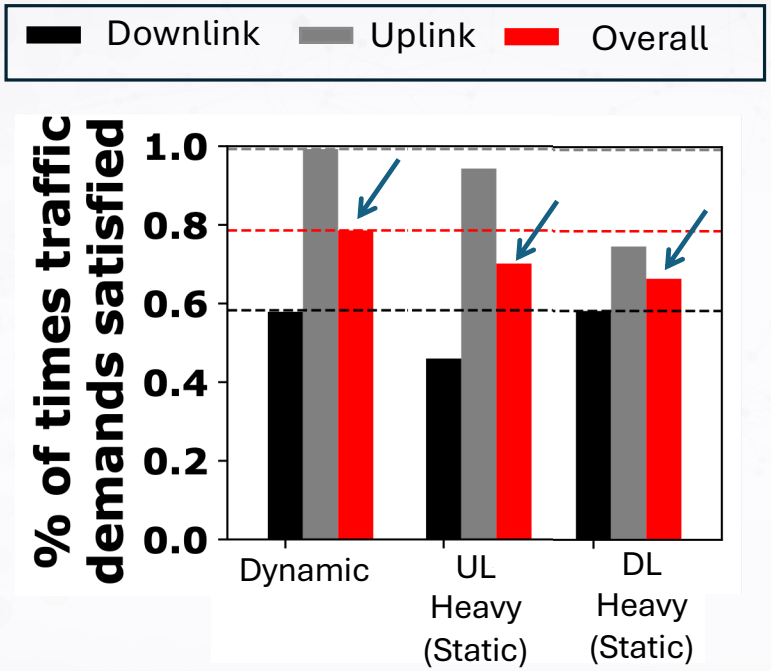
Proposed/Modified modules

Real world measurement data

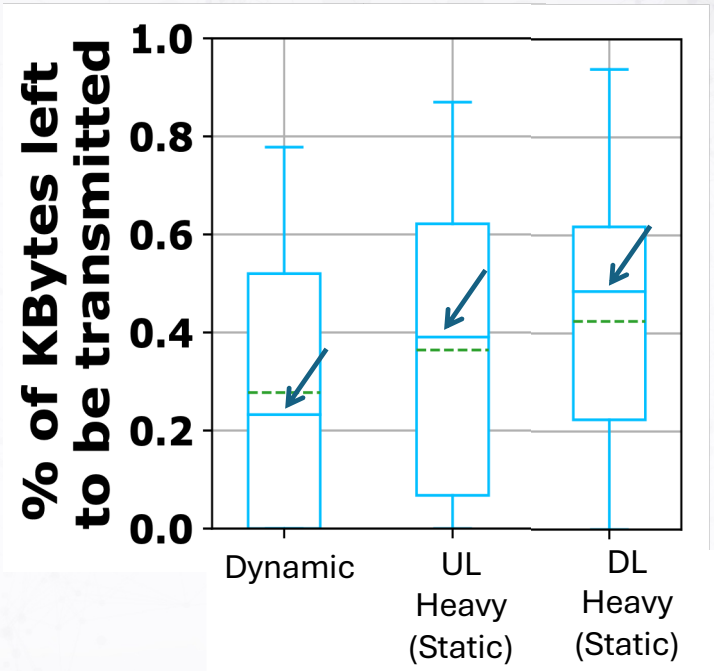
Prototype implemented on an 3GPP (OAI) compliant testbed[1]

[1]*M. Ghoshal, S. Mohanti and D. Koutsonikolas, "Enabling Emerging Applications in 5G Through UE-Assisted Proactive PHY Frame Configuration," 2024 IEEE 35th International Symposium on Personal, Indoor and Mobile Radio Communications (PIMRC)*

Results



Dynamic allocation satisfies the overall traffic demands more times than any other static configurations



With dynamic allocation users are left with the lowest % of un-transmitted data

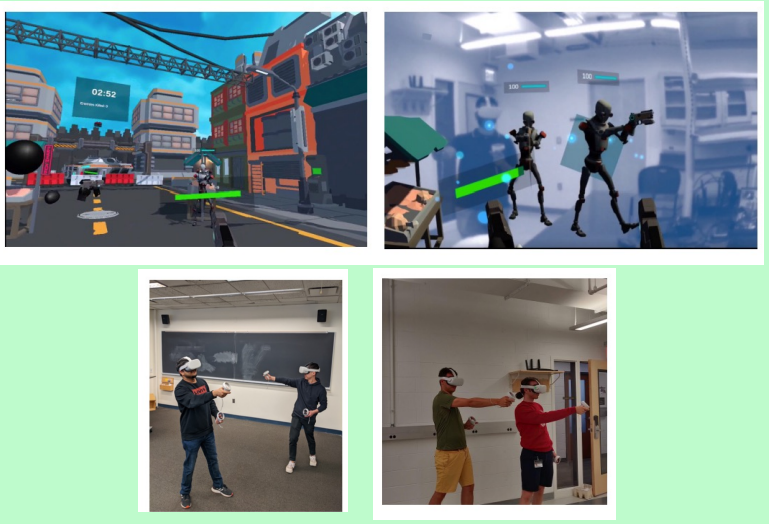
Other Works

Extensive work with 60 GHz WLANs (802.11ad)

- Network and application performance, energy efficiency, and interference characterization *(IEEE INFOCOM 2021, Elsevier ComCom)*
- Enabled and open-sourced Multi-Path TCP (MPTCP) on dual-band 60 GHz/5 GHz smartphones *(ACM WinTech 2021)*
- Network throughput predictability for 60 GHz WLANs *(PAM 2021)*
- Intelligent link adaptation framework for 60 GHz WLANs *(ACM CoNEXT 2021)*

Human Computer Interaction (HCI) work

- ***"Escape from Kyle-Earth"*** – a multi-player immersive VR/AR game designed to understand the effects of co-located gaming on user experience *(IEEE Conference on Games 2022)*



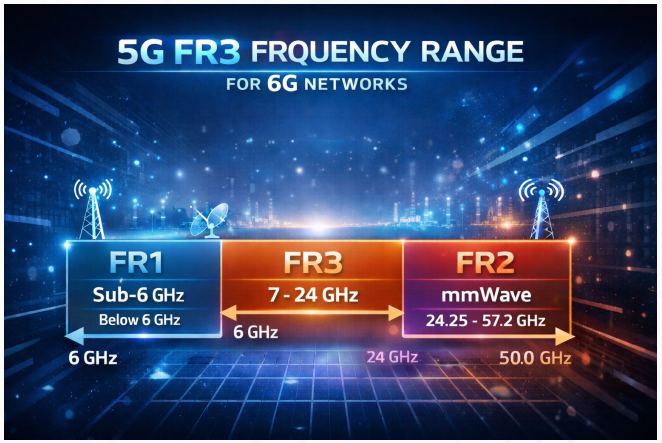
Ongoing and Future Work

Continuing Large Scale Measurement Studies

- Future work → 5G is still evolving, 6G in the horizon







Integration of Terrestrial and Non-Terrestrial Networks

- Critical step towards achieving seamless connectivity in 6G
  - 3GPP Rel. 17, 18, 19, 20
  - O-RAN NextG Research Group Roadmap

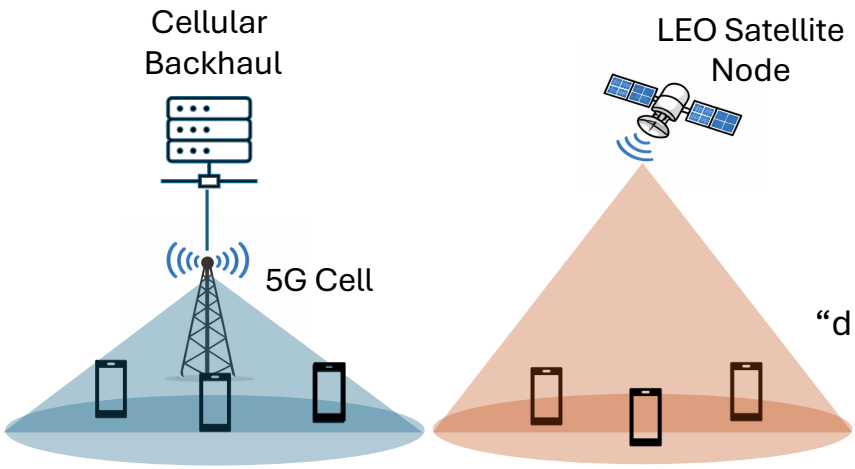
- LEO satellite networks and cellular networks have very different characteristics
  - Complementary to each other?

Integration of Terrestrial and Non-Terrestrial Networks

- Ongoing work: **Integration at the transport layer** via MPTCP/MPQUIC
  - Use both networks to boost throughput
  - Select best network based on network conditions

Integration of Terrestrial and Non-Terrestrial Networks

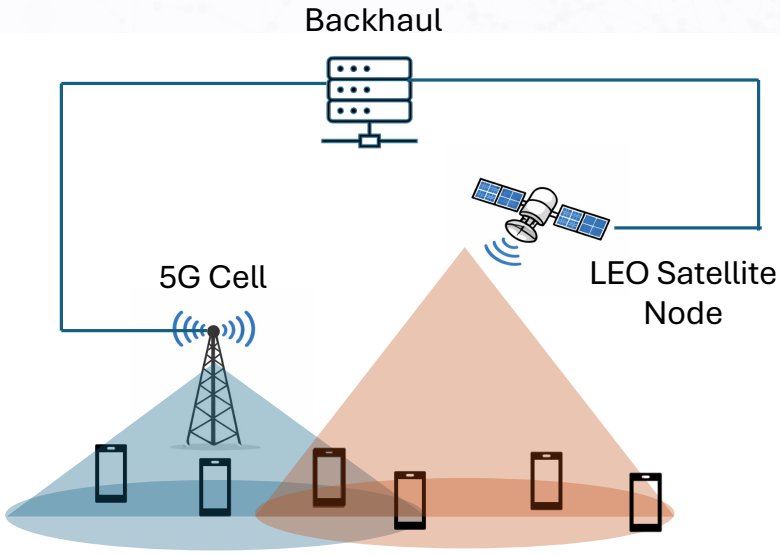
- Ongoing work: **Integration at the transport layer** via MPTCP/MPQUIC
  - Use both networks to boost throughput
  - Select best network based on network conditions



Currently NTNs act as a
“disconnected layer” to existing TNs

Integration of Terrestrial and Non-Terrestrial Networks

- Ongoing work: **Integration at the transport layer** via MPTCP/MPQUIC
  - Use both networks to boost throughput
  - Select best network based on network conditions



Challenges

- Managing different
  in single unified fra
  difficult
  - Different RTTs
  - Higher dopple
  - Frequent hand

6G envisions unification of both TNs and NTNs in a single and harmonized standard

Need for common control layer → TN-NTN mobility management, QoS-aware unified resource management

Integration of Terrestrial and Non-Terrestrial Networks

Existing Academic NTN Research Platforms

- Network simulators/emulators
  - Hypatia
  - LeoEM
  - OpenSN
- Channel/Propagation Models
  - GR-LEO
  - RF Emulators
- Trajectory/Orbit Models
  - SGP4
  - Skyfield

X *Isolated tools across layers*
X *No cross-layer control*
X *No joint TN-NTN modelling*
X *Limited application evaluation*

Industry Solutions

- Few companies have partial to complete TN-NTN testing solutions

KEYSIGHT

CelesS

VIAVI Solutions

X *Closed and proprietary*
X *Expensive*

Towards an Open Unified TN-NTN Research Platform

✓ Cross-layer integration
✓ Joint TN-NTN operation
✓ Open and programmable architecture
✓ Hardware-in-the-loop
✓ End-to-end application evaluation
✓ AI-driven optimization

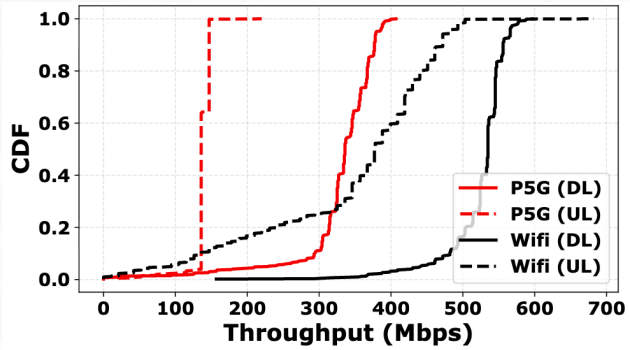
Exploration of Private 5G in Enterprise Settings and Its Convergence with Wi-Fi

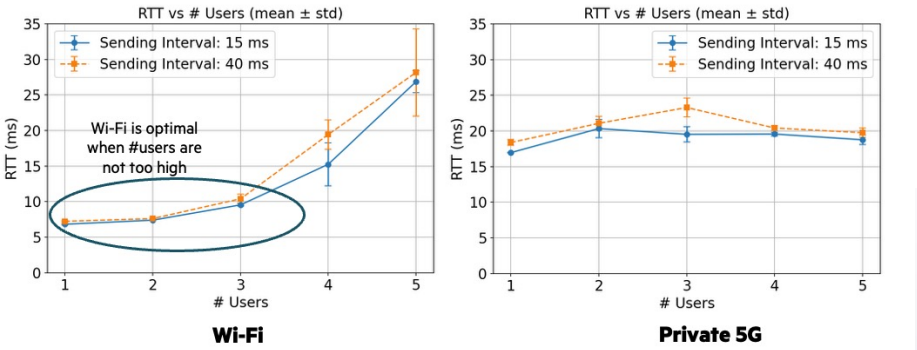
- Global private 5G market share
  - 2023: $2 billion USD
  - Projected by 2030: $36 billion USD

- P5G benefits over WiFi
  - Scheduled access vs. WiFi's random channel access
    - Guaranteed latency even when network load increases
  - Slicing in P5G allows customizable network resource allocation for multiple users under the same network
  - Fine tunable parameters based on network requirements
  - Stronger security mechanisms

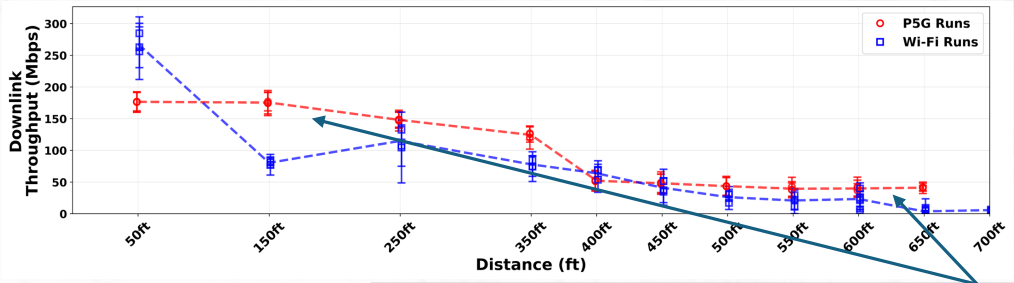
Exploration of Private 5G in Enterprise Settings and Its Convergence with Wi-Fi

WiFi higher capacity at short distances

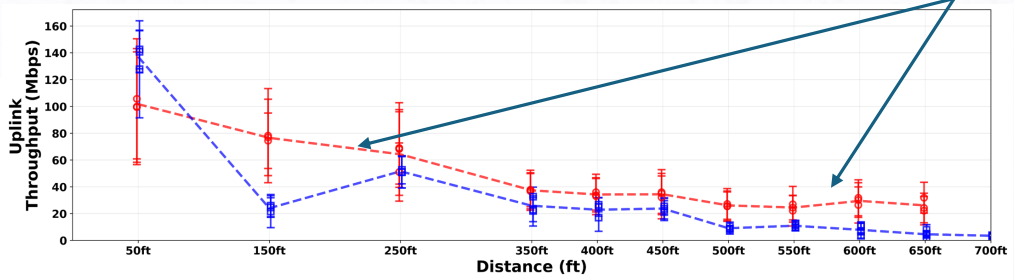




Best of both worlds?
Unify P5G and WiFi



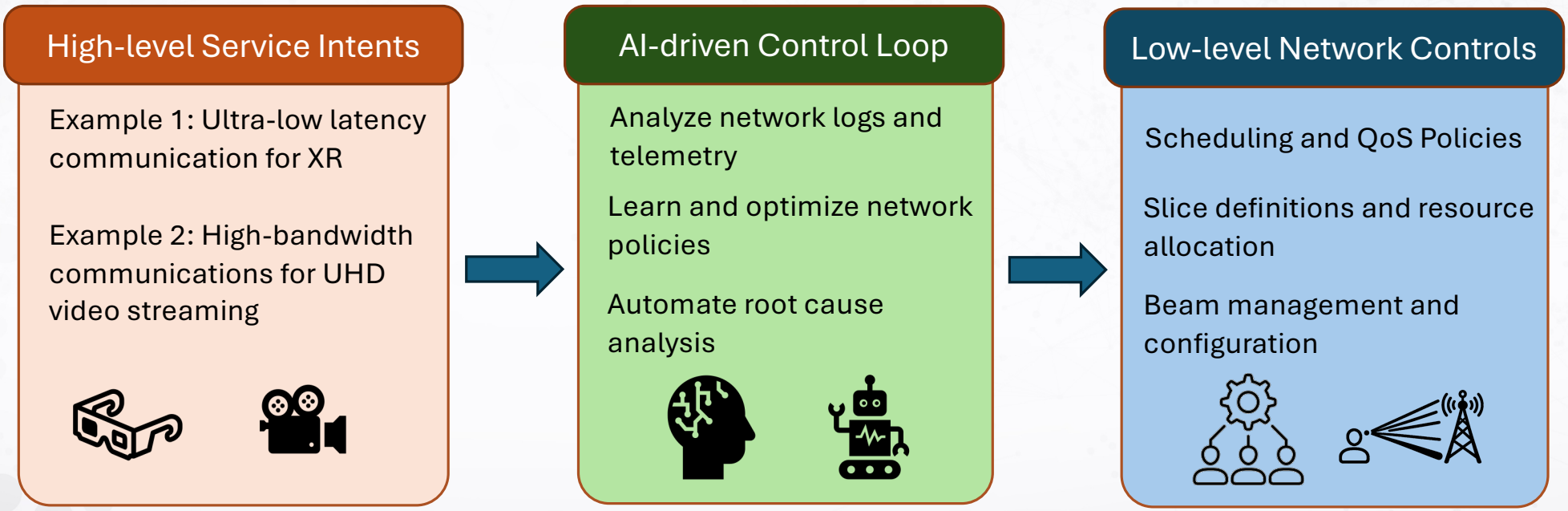
P5G more resilient with longer distances



"Beyond Wi-Fi: Private 5G for better Multi-user Quality of Experience", Ghosh et. al., HotMobile 2026

Intent-Driven Networking with AI-Augmented Policy Understanding

Designing AI-driven systems that align network behavior with high-level service intents



Focus: Leveraging AI and telemetry data to create resilient, autonomous, and user-aware networks

Awards and Recognition

- 1 patent accepted
- 12 datasets open sourced; more coming
  - Best dataset award (PAM'21); Community contribution award runner up (CoNEXT'25)
- Presented 5G Standalone results to T-Mobile
- Invited talks: RPI (2026), HPE (2025), New England Systems Day (2025), Juniper Networks (2024)
- Northeastern University College of Engineering Outstanding Research Award

Industry Collaborations

Hewlett Packard Enterprise

Rakuten

interdigital.

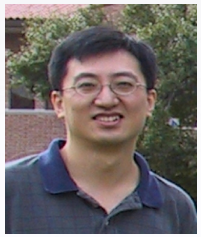
FUTUREWEI Technologies

AT&T

OOKLA

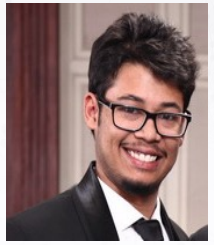
Acknowledgement

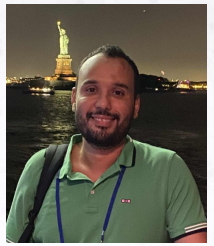

Dimitrios Koutsonikolas


Y. Charlie Hu


Joerg Widmer


Puneet Sharma


Shivang Aggarwal


Mohamed Ibrahim


Subhramoy Mohanti



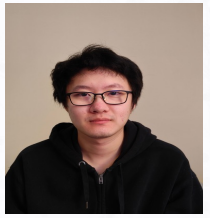
Abbas Kiani


Eduardo Baena


Imran Khan


Phuc Dinh

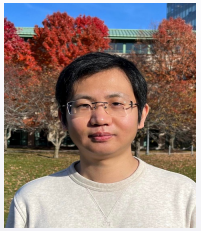

Sizhe Wang


Yufei Feng


Zhekun Yu

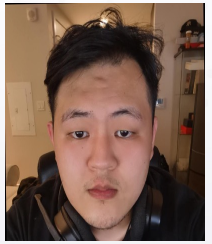

Z. Jonny Kong


Omar Basit


Qiang Xu


Zihan Xu


Fukun Chen


Zixiao Lu


Piyali Banerjee

Thank You!

Email: ghoshal.m@northeastern.edu

QR for all 5G/xG Research Papers, Code, and Open Datasets